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(54) **SUMMARY PAGE GENERATION USING DOCUMENTS**

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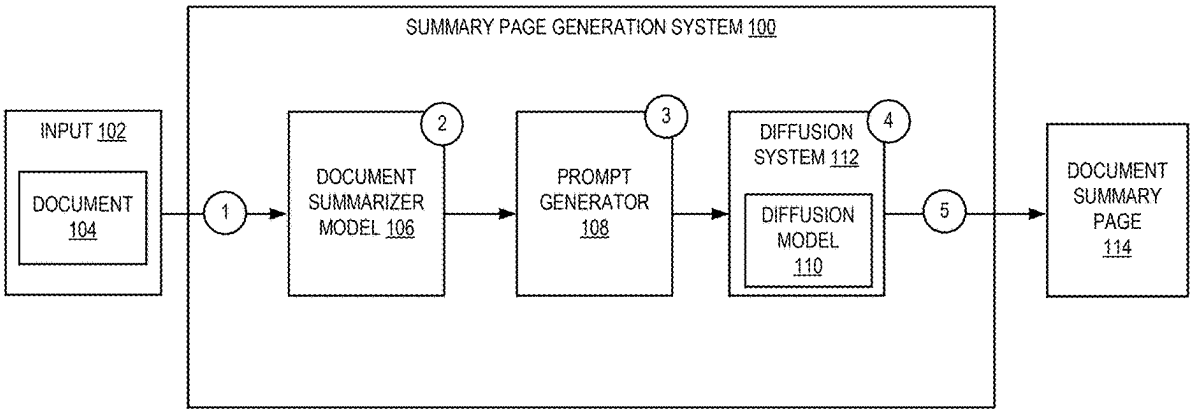
(57) **ABSTRACT**

Embodiments are disclosed for summary page generation using a document. The method may include receiving a text document. The method may further include generating a test summary based on the text document and a structured representation of the text summary using the document summarized model. The method may further include generating an image generation prompt based on the text summary and the structured representation of the text summary using a prompt generator. The method may further include generating a multimedia summary document corresponding to the text document using a diffusion model and the image generation prompt. The multimedia summary document includes a generated background imagery based on the text summary. The multimedia summary document includes at least a portion of the text summary which is placed within the multimedia summary document based on the structed representation of the text summary.



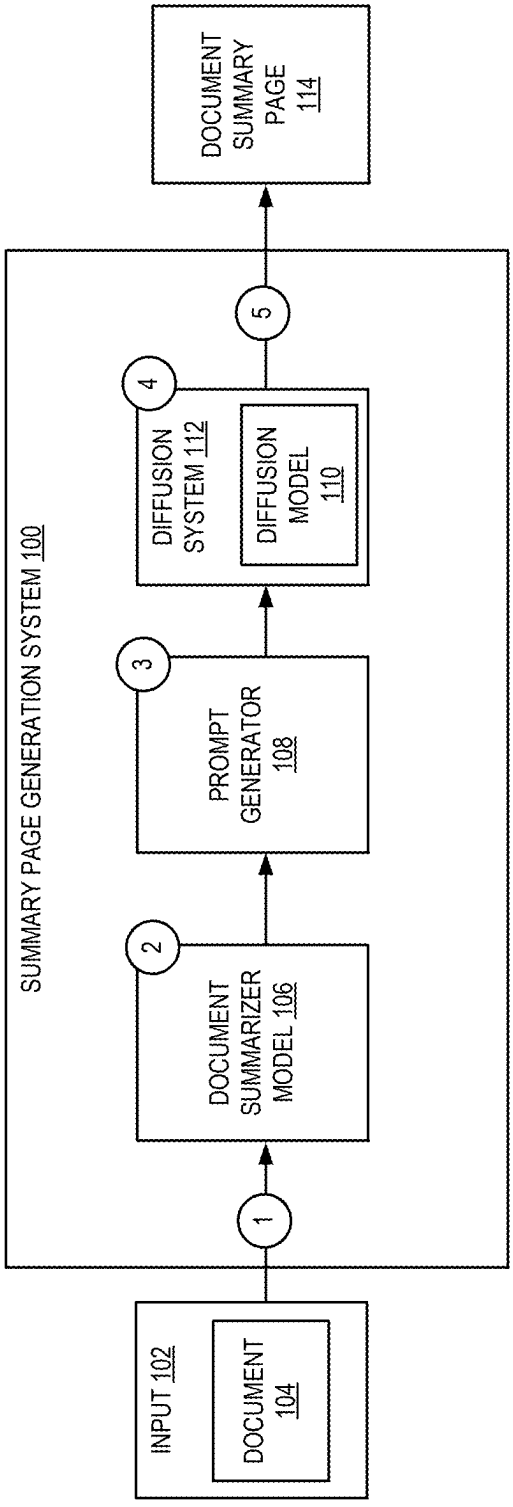

INPUT 102
DOCUMENT 104
SUMMARY PAGE GENERATION SYSTEM 100
1
DOCUMENT SUMMARIZER MODEL 106
2
PROMPT GENERATOR 108
3
DIFFUSION SYSTEM 112
DIFFUSION MODEL 110
4
5
DOCUMENT SUMMARY PAGE 114

FIG. 1

202

```
<html>
    <head>
        <title>ChatGPT: Optimizing Language Models for Dialogue</title>
    </head>
    <body>
        <h1>ChatGPT: Optimizing Language Models for Dialogue</h1>
        <p>Unlock conversation potential with ChatGPT. Optimize dialogue for better engagement.</p>
    </body>
</html>
```

204

```
<html>
  <head>
    <title> A Peer-to-Peer Electronic  System</title>
  </head>
  <body>
    <h1> Electronic Cash System</h1>
    <p>By: ES  </p>
  </body>
</html>
```

FIG. 2

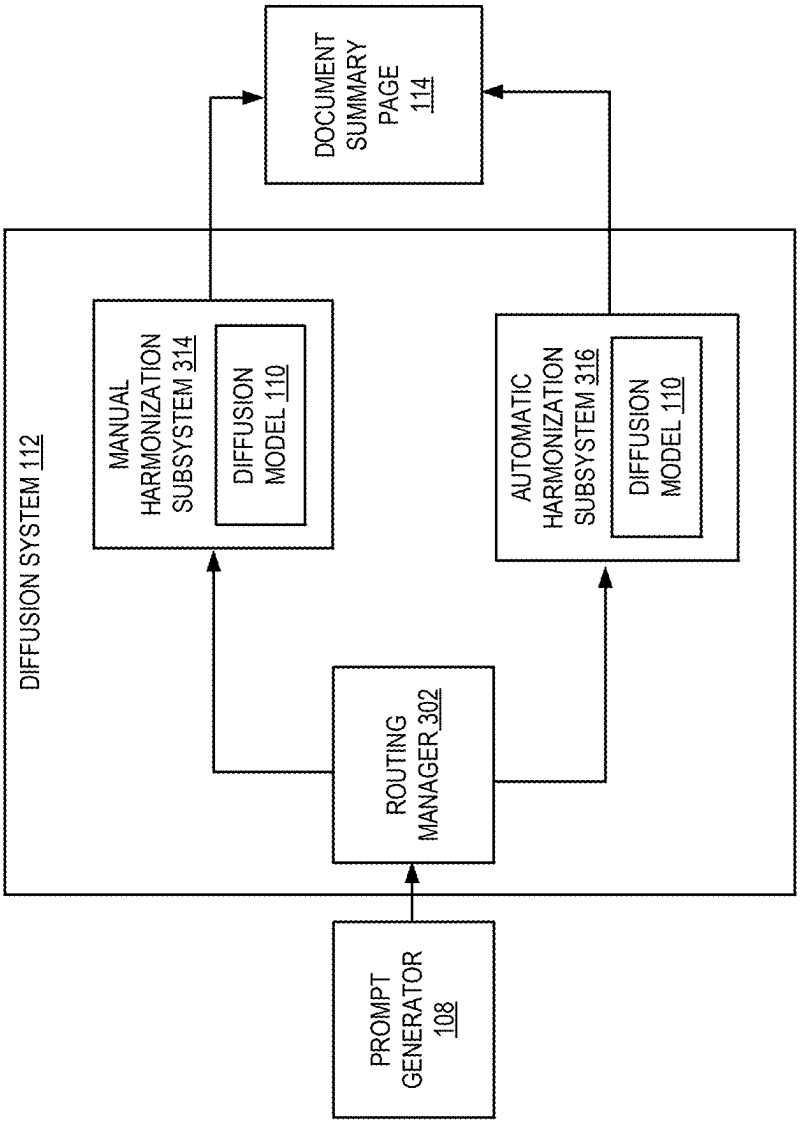

DOCUMENT SUMMARY PAGE 114
DIFFUSION SYSTEM 112
MANUAL HARMONIZATION SUBSYSTEM 314
DIFFUSION MODEL 110
AUTOMATIC HARMONIZATION SUBSYSTEM 316
DIFFUSION MODEL 110
ROUTING MANAGER 302
PROMPT GENERATOR 108

FIG. 3

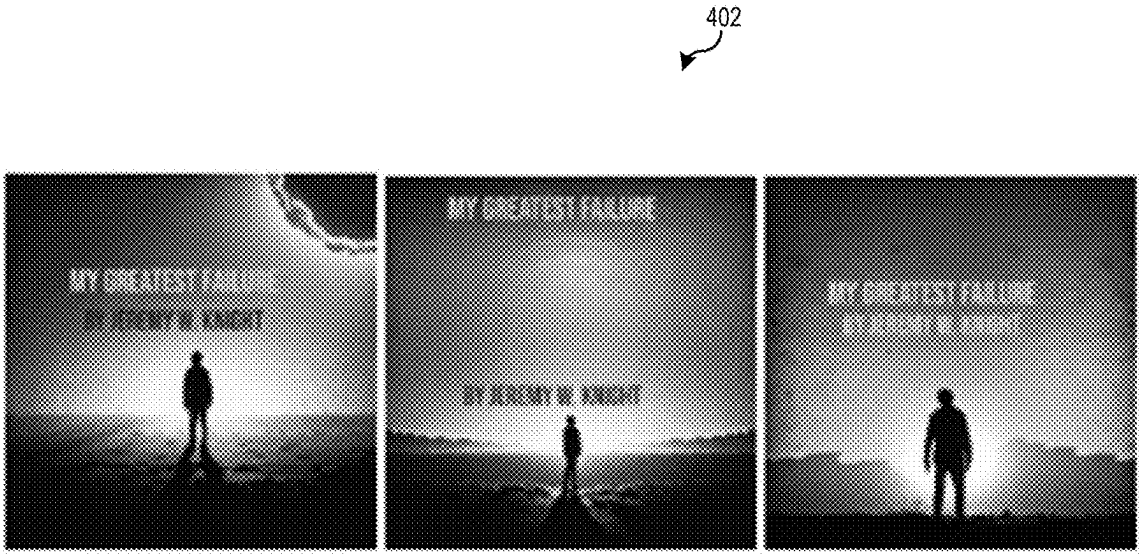

402
404
My Greatest Failure
My Greatest Failure
My Greatest Failure

FIG. 4

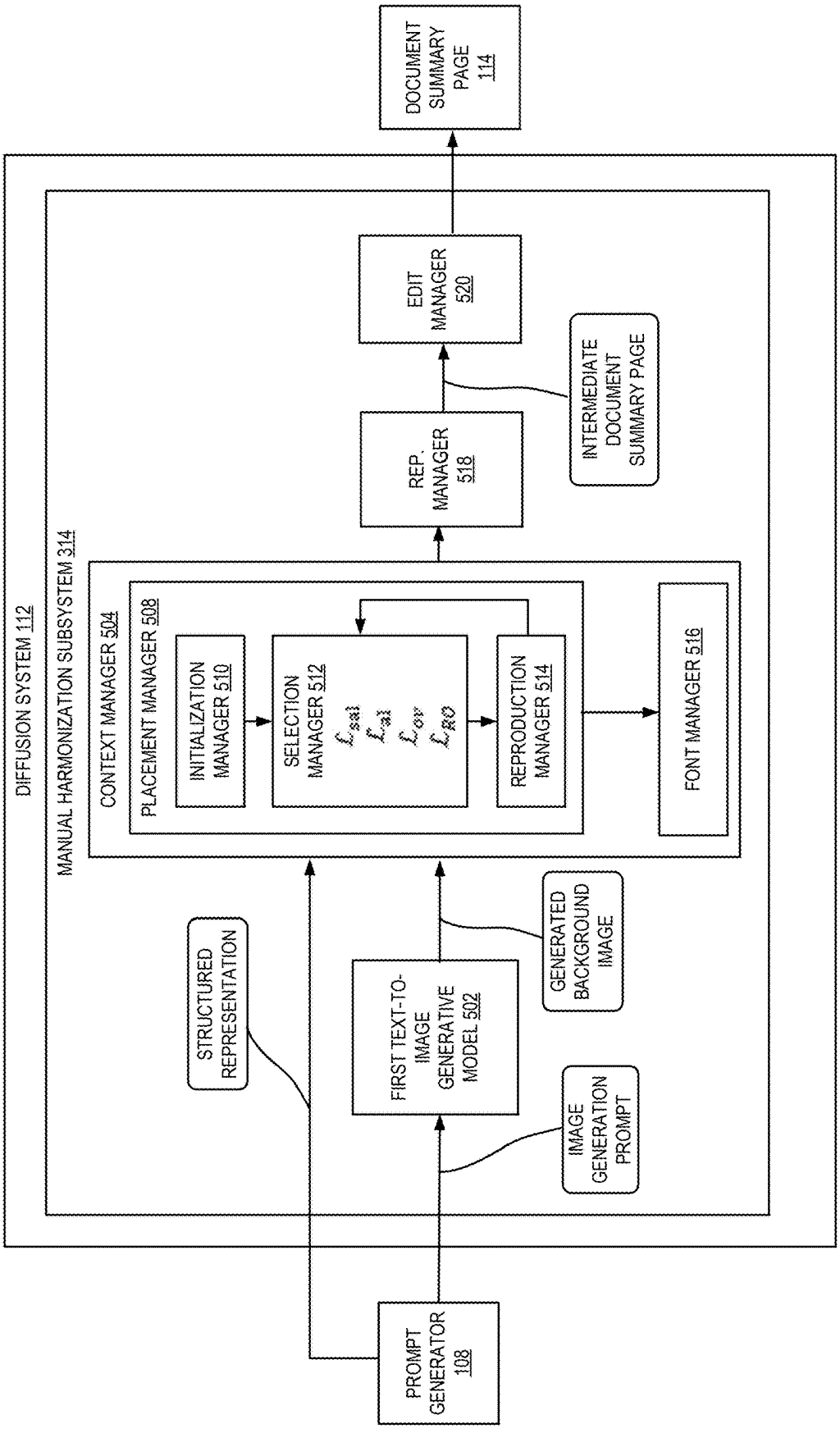

DOCUMENT SUMMARY PAGE 114
DIFFUSION SYSTEM 112
MANUAL HARMONIZATION SUBSYSTEM 314
EDIT MANAGER 520
REP. MANAGER 518
INTERMEDIATE DOCUMENT SUMMARY PAGE
CONTEXT MANAGER 504
PLACEMENT MANAGER 508
INITIALIZATION MANAGER 510
SELECTION MANAGER 512
$\mathcal{L}_{sal}$
$\mathcal{L}_{al}$
$\mathcal{L}_{ov}$
$\mathcal{L}_{RO}$
REPRODUCTION MANAGER 514
FONT MANAGER 516
STRUCTURED REPRESENTATION
FIRST TEXT-TO-IMAGE GENERATIVE MODEL 502
GENERATED BACKGROUND IMAGE
IMAGE GENERATION PROMPT
PROMPT GENERATOR 108

FIG. 5

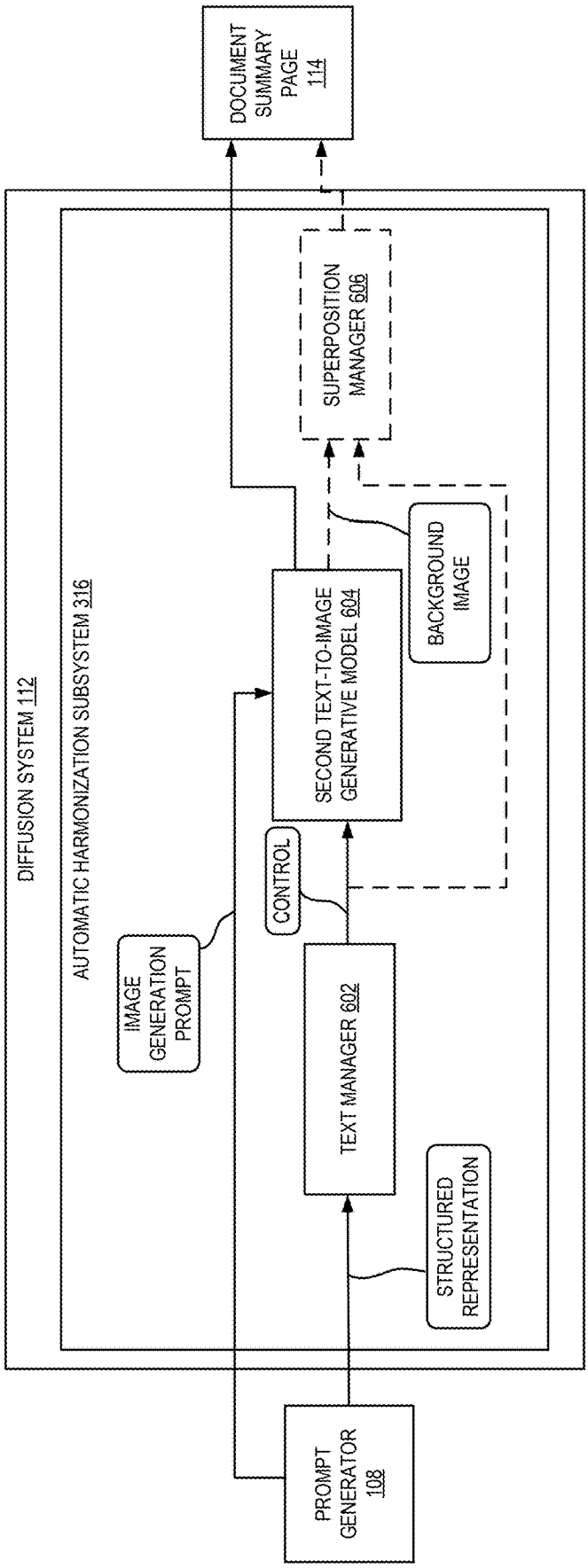

DOCUMENT SUMMARY PAGE 114
DIFFUSION SYSTEM 112
AUTOMATIC HARMONIZATION SUBSYSTEM 316
SUPERPOSITION MANAGER 606
BACKGROUND IMAGE
SECOND TEXT-TO-IMAGE GENERATIVE MODEL 604
CONTROL
IMAGE GENERATION PROMPT
TEXT MANAGER 602
STRUCTURED REPRESENTATION
PROMPT GENERATOR 108

FIG. 6

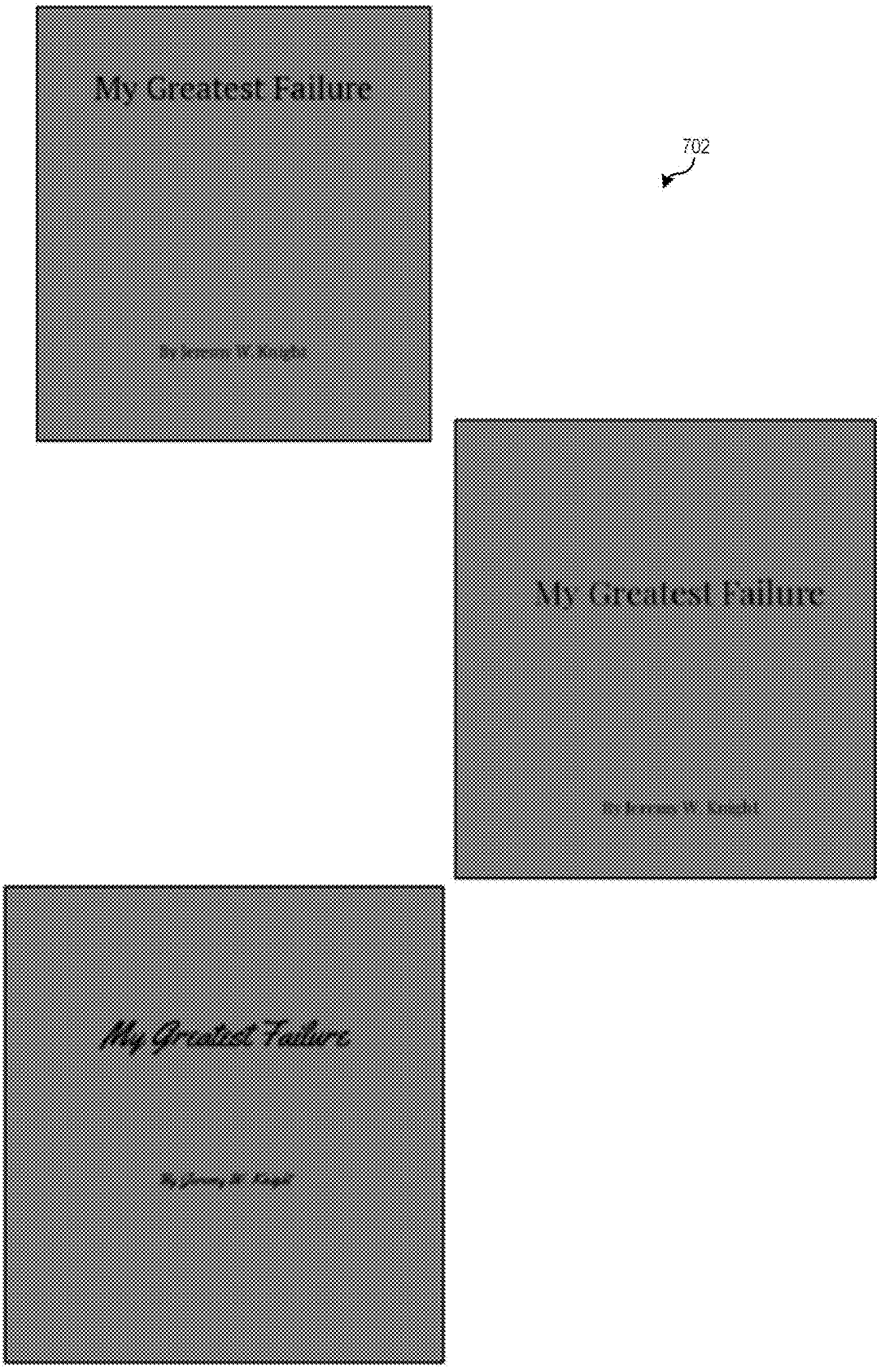

My Greatest Failure
702
My Greatest Failure
My Greatest Failure

FIG. 7

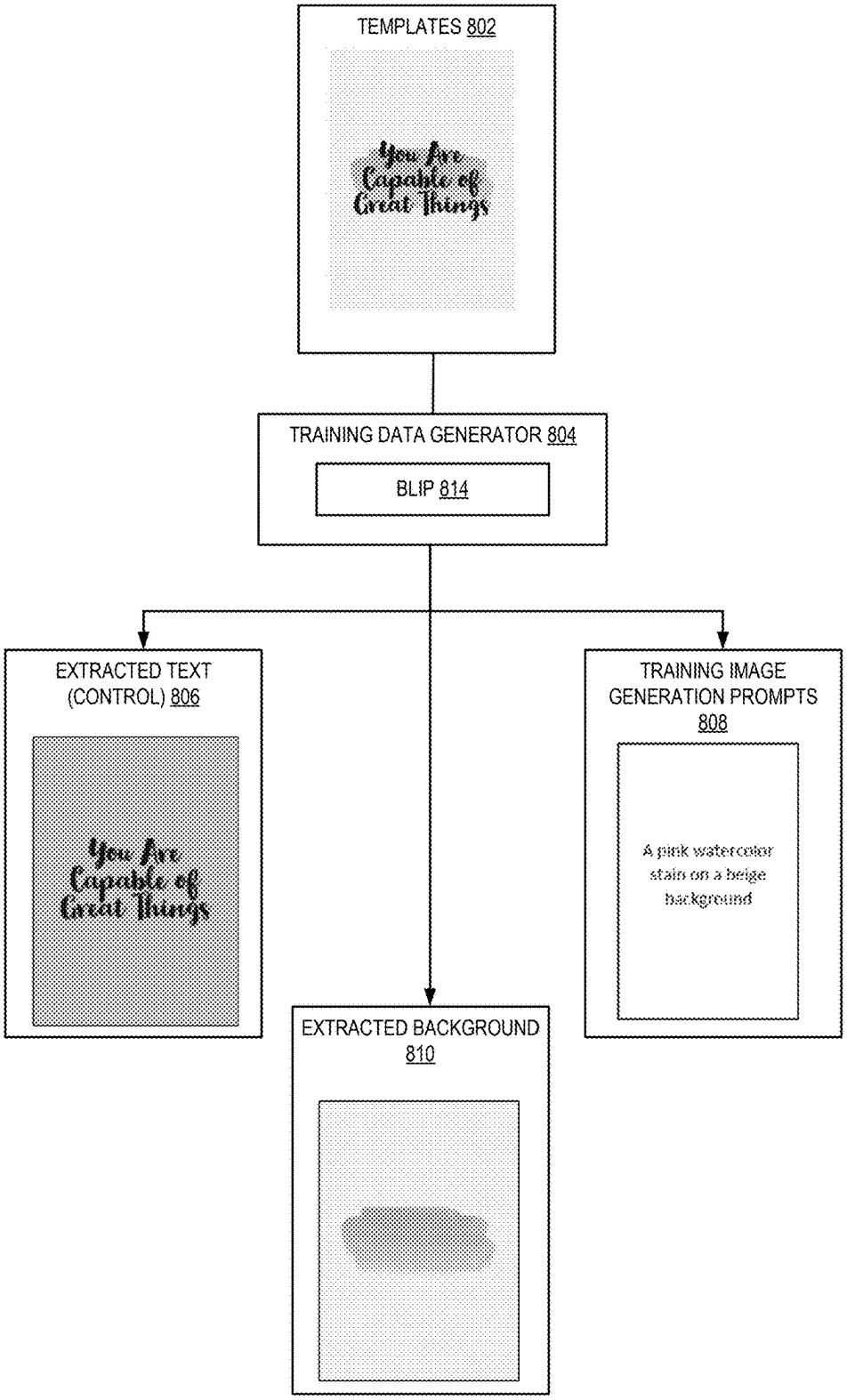

TEMPLATES 802
You Are Capable of Great Things
TRAINING DATA GENERATOR 804
BLIP 814
EXTRACTED TEXT (CONTROL) 806
You Are Capable of Great Things
TRAINING IMAGE GENERATION PROMPTS 808
A pink watercolor stain on a beige background
EXTRACTED BACKGROUND 810

FIG. 8

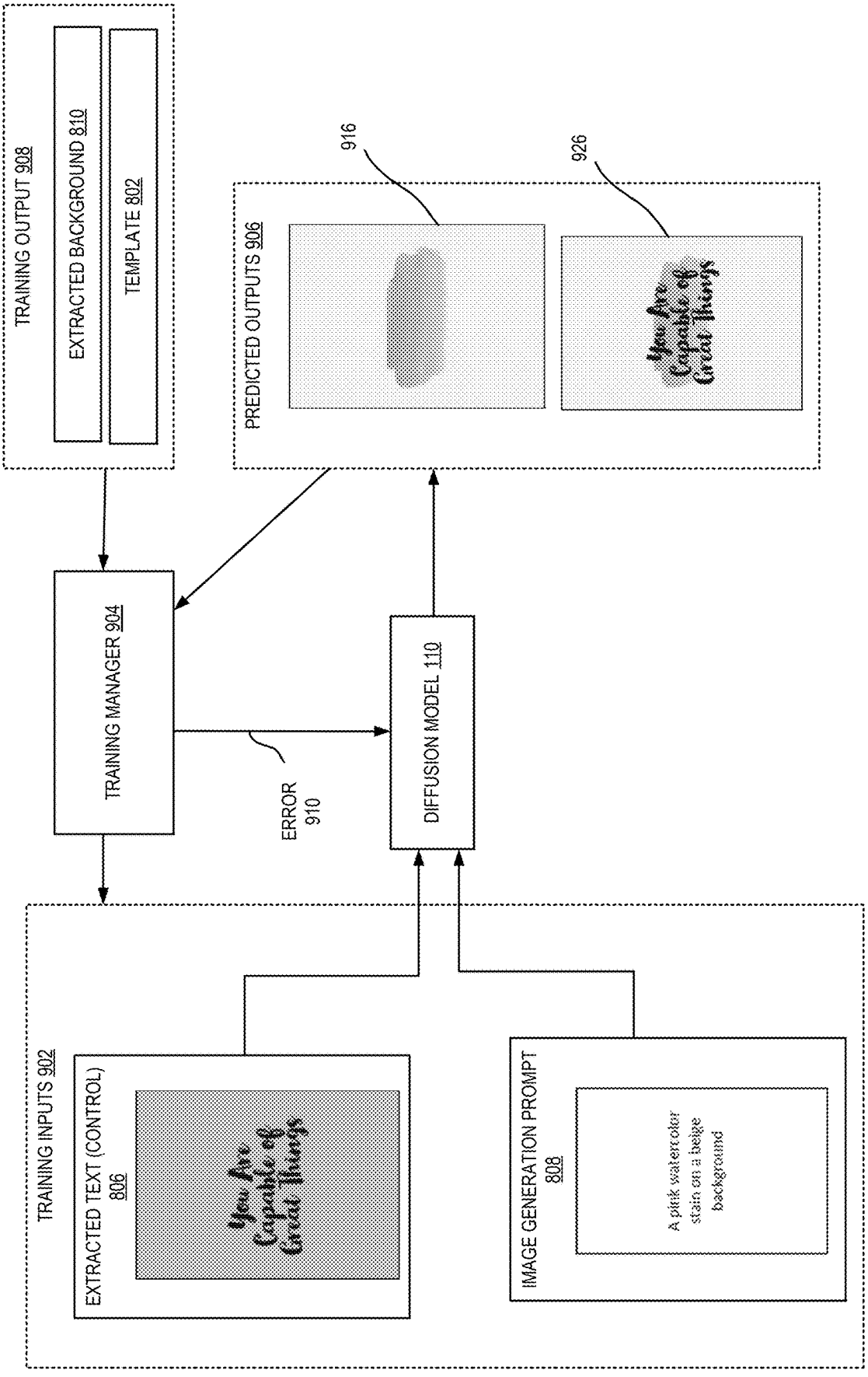

TRAINING OUTPUT 908
EXTRACTED BACKGROUND 810
TEMPLATE 802
PREDICTED OUTPUTS 906
916
926
You Are Capable of Great Things
TRAINING MANAGER 904
DIFFUSION MODEL 110
ERROR 910
TRAINING INPUTS 902
EXTRACTED TEXT (CONTROL) 806
You Are Capable of Great Things
IMAGE GENERATION PROMPT 808
A pink watercolor stain on a beige background

FIG. 9

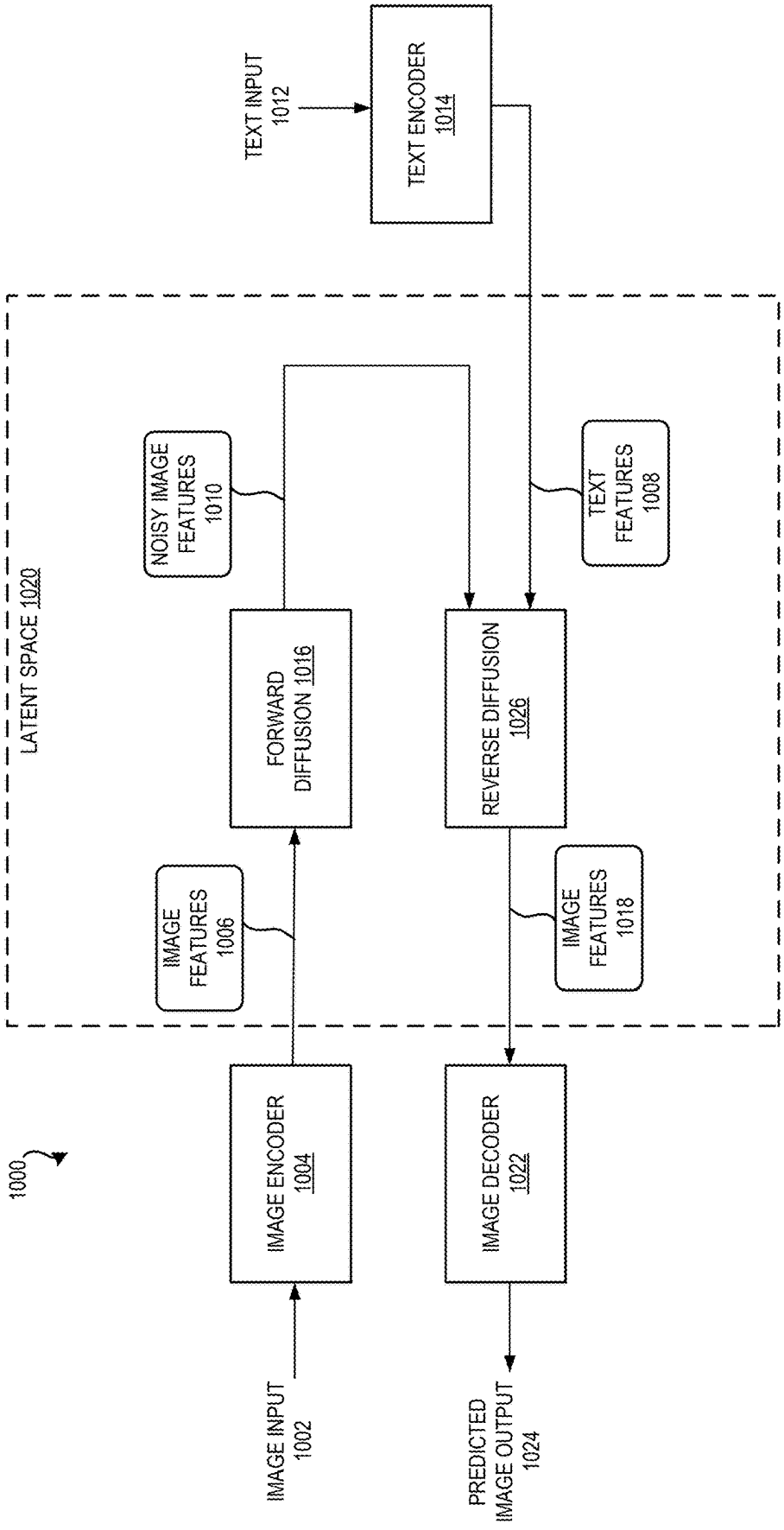

1000
IMAGE INPUT
1002
IMAGE ENCODER
1004
IMAGE DECODER
1022
PREDICTED
IMAGE OUTPUT
1024
LATENT SPACE 1020
IMAGE
FEATURES
1006
FORWARD
DIFFUSION 1016
REVERSE DIFFUSION
1026
IMAGE
FEATURES
1018
NOISY IMAGE
FEATURES
1010
TEXT
FEATURES
1008
TEXT INPUT
1012
TEXT ENCODER
1014

FIG. 10

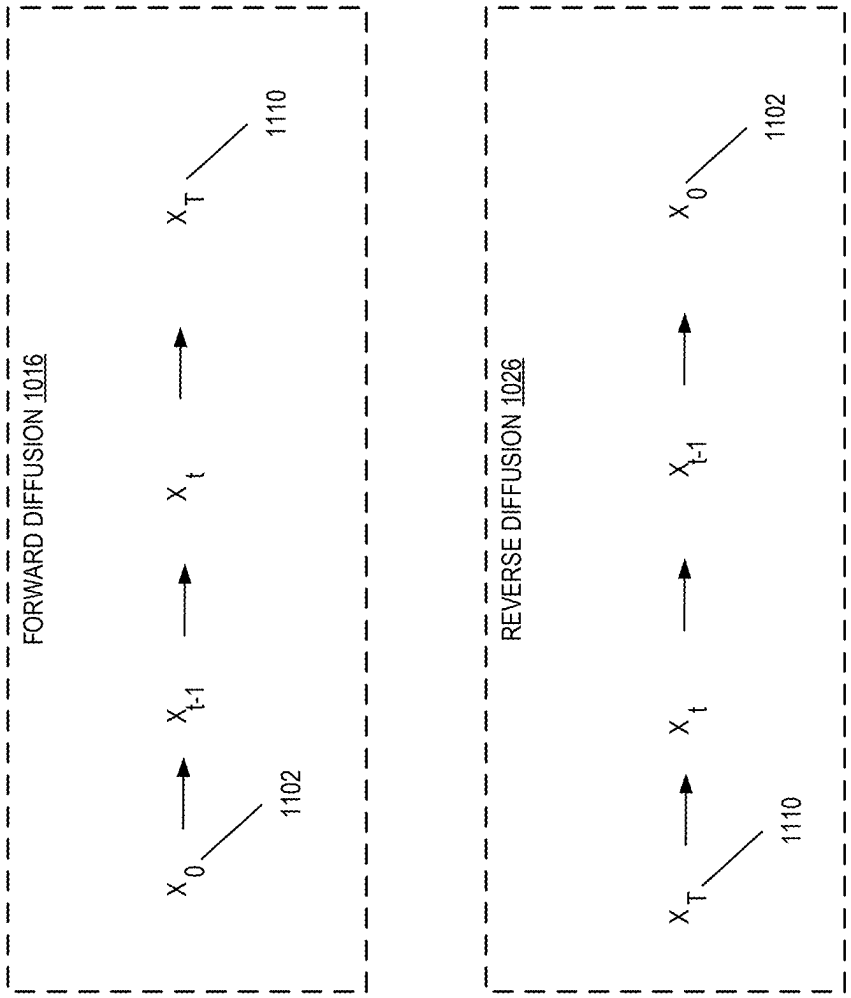

FORWARD DIFFUSION 1016
X_0
1102
X_{t-1}
X_t
X_T
1110
REVERSE DIFFUSION 1026
X_T
1110
X_t
X_{t-1}
X_0
1102

FIG. 11

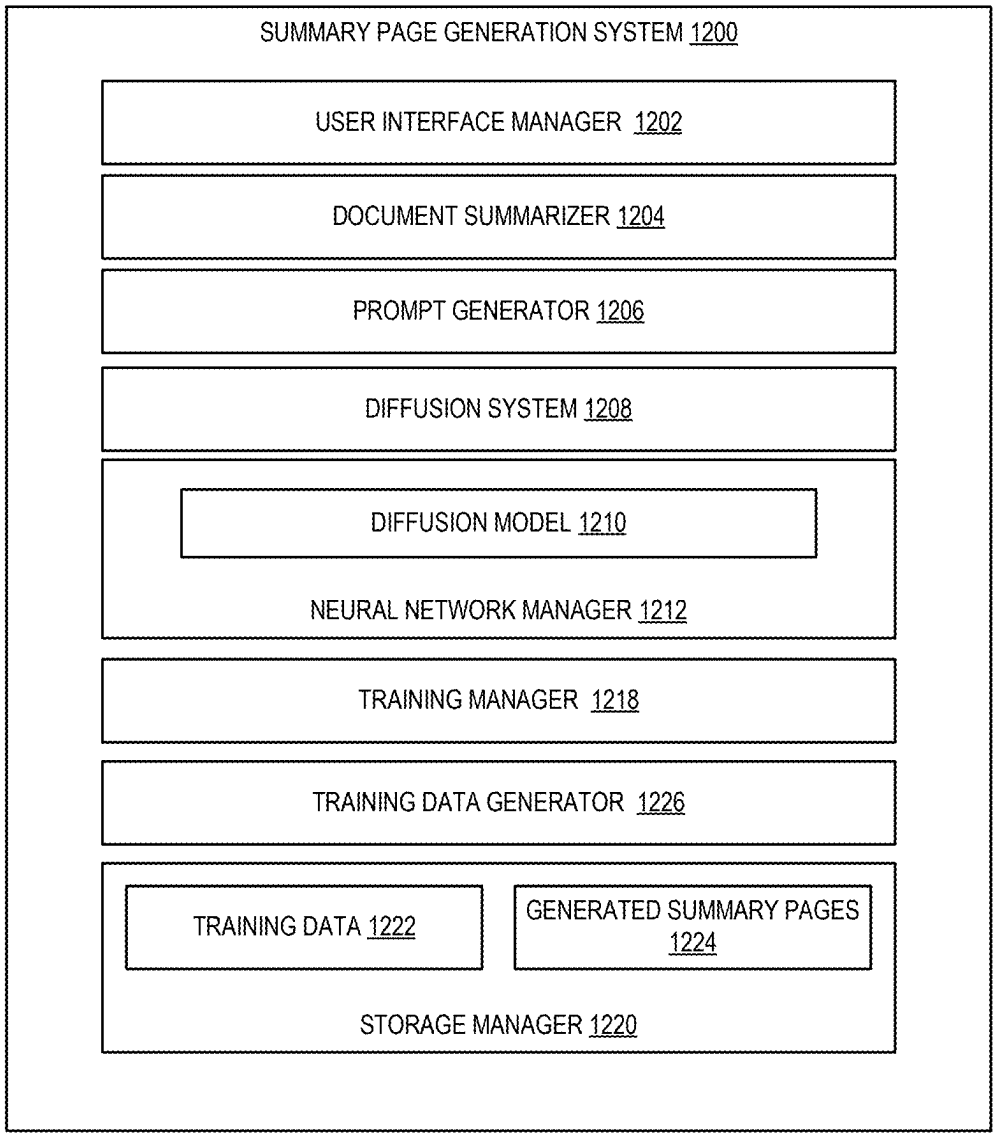

SUMMARY PAGE GENERATION SYSTEM 1200
USER INTERFACE MANAGER 1202
DOCUMENT SUMMARIZER 1204
PROMPT GENERATOR 1206
DIFFUSION SYSTEM 1208
DIFFUSION MODEL 1210
NEURAL NETWORK MANAGER 1212
TRAINING MANAGER 1218
TRAINING DATA GENERATOR 1226
TRAINING DATA 1222
GENERATED SUMMARY PAGES 1224
STORAGE MANAGER 1220

FIG. 12

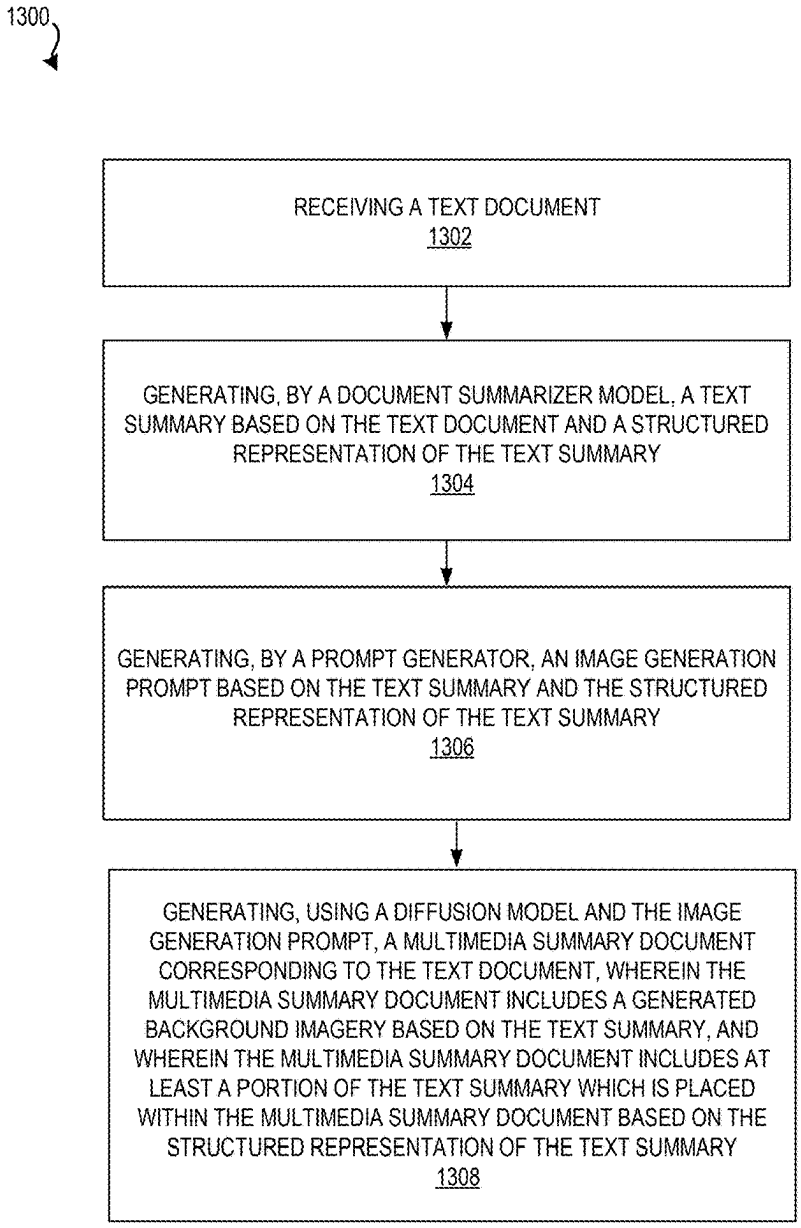

1300
RECEIVING A TEXT DOCUMENT
1302
GENERATING, BY A DOCUMENT SUMMARIZER MODEL, A TEXT SUMMARY BASED ON THE TEXT DOCUMENT AND A STRUCTURED REPRESENTATION OF THE TEXT SUMMARY
1304
GENERATING, BY A PROMPT GENERATOR, AN IMAGE GENERATION PROMPT BASED ON THE TEXT SUMMARY AND THE STRUCTURED REPRESENTATION OF THE TEXT SUMMARY
1306
GENERATING, USING A DIFFUSION MODEL AND THE IMAGE GENERATION PROMPT, A MULTIMEDIA SUMMARY DOCUMENT CORRESPONDING TO THE TEXT DOCUMENT, WHEREIN THE MULTIMEDIA SUMMARY DOCUMENT INCLUDES A GENERATED BACKGROUND IMAGERY BASED ON THE TEXT SUMMARY, AND WHEREIN THE MULTIMEDIA SUMMARY DOCUMENT INCLUDES AT LEAST A PORTION OF THE TEXT SUMMARY WHICH IS PLACED WITHIN THE MULTIMEDIA SUMMARY DOCUMENT BASED ON THE STRUCTURED REPRESENTATION OF THE TEXT SUMMARY
1308

FIG. 13

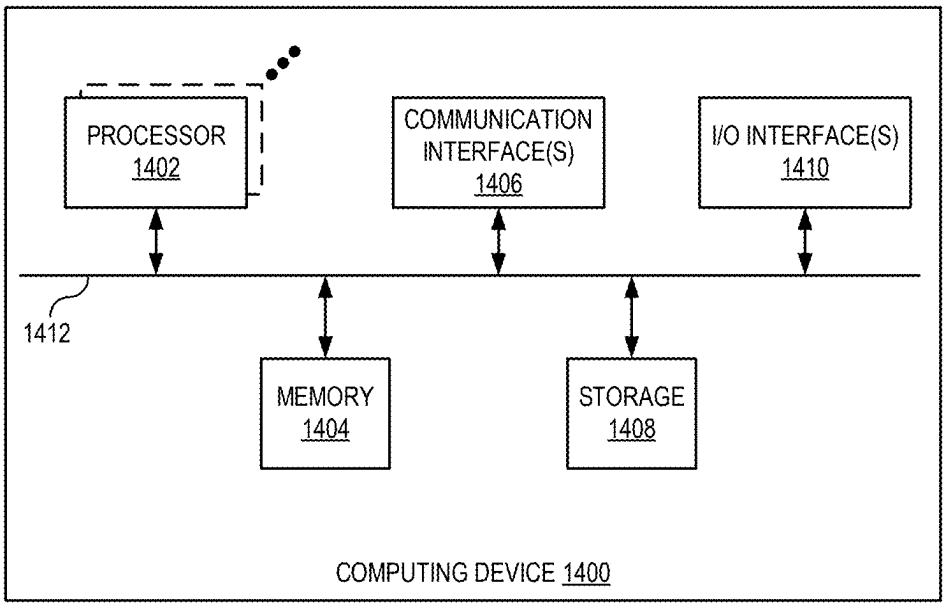

PROCESSOR
1402
COMMUNICATION INTERFACE(S)
1406
I/O INTERFACE(S)
1410
1412
MEMORY
1404
STORAGE
1408
COMPUTING DEVICE 1400

FIG. 14

SUMMARY PAGE GENERATION USING DOCUMENTS

BACKGROUND

[0001] Digital tools allow artists to manifest creative efforts in a digital workspace. For example, an artist (or other creator) can create a summary page for a document using the digital workspace. The summary page can include a synopsis of the content of the document, set the tone for the reader, capture the reader's attention, and/or convey a message of the document using text and images alike.

SUMMARY

[0002] Introduced here are techniques/technologies that generate a multimedia summary page (e.g., a cover page, a poster, an infographic, a flyer, etc.) based on a document (e.g., a report, proposal, assignment, etc.). The summary page generation system described herein leverages the content of the document to determine information about the document such as the title of the document, a subtitle of the document, an author of the document, and a summary of the document. The information is arranged on a generated background relevant to the theme of the document in a visually aesthetic composition. The output of the summary page generation system is a multimedia summary page that includes generated background imagery and a text description of the document.

[0003] More specifically, in one or more embodiments, the summary page generation system uses a Socratic Model framework to pass information between language models and text-to-image generative models, bridging the domain gap between visual and text features. The summary page generation system generates a diverse set of summary pages using two degrees of generated content. The first degree of generated content includes a language model determining a structure of extracted text context from the document and generating a prompt for a text-to-image generative model. The generated prompt determined using the language model encourages diversity because different generated prompts can be determined for the same input document. The second degree of generated content includes using the text-to-image generative model to generate an image using the generated prompt. The generated image determined using the text-to-image generative models encourages diversity because different generated images can be determined for the same prompt.

[0004] The summary page generation system of the present disclosure includes multiple pipelines used to generate multimedia summary pages, where any given pipeline can be executed to generate a diverse set of summary pages. For example, in some embodiments, a first pipeline iteratively harmonizes a text layout by optimizing the position of the text, a style of font, a size of the text, and a color of the text on generated background imagery, and a second pipeline automatically harmonizes a background given a text layout using generative artificial intelligence (AI) such as a diffusion model.

[0005] Additional features and advantages of exemplary embodiments of the present disclosure will be set forth in the description which follows, and in part will be obvious from the description, or may be learned by the practice of such exemplary embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The detailed description is described with reference to the accompanying drawings in which:

[0007] FIG. **1** illustrates a diagram of a process of generating summary pages based on documents, in accordance with one or more embodiments;

[0008] FIG. **2** illustrates two examples of structured representations of the summary page content generated by a language model, in accordance with one or more embodiments;

[0009] FIG. **3** illustrates a diagram of a process of the diffusion system, in accordance with one or more embodiments;

[0010] FIG. **4** illustrates examples of diverse multimedia summary documents determined using the summary page generation system, in accordance with one or more embodiments;

[0011] FIG. **5** illustrates a diagram of a process of the manual harmonization subsystem, in accordance with one or more embodiments;

[0012] FIG. **6** illustrates a diagram of a process of the automatic harmonization subsystem, in accordance with one or more embodiments;

[0013] FIG. **7** illustrates three examples of text canvases (or controls) determined using the text manager, in accordance with one or more embodiments;

[0014] FIG. **8** illustrates an example process of generating training data to fine-tune a diffusion model, in accordance with one or more embodiments;

[0015] FIG. **9** illustrates an example process of training a text-to-image generative model, in accordance with one or more embodiments;

[0016] FIG. **10** illustrates an example implementation of a diffusion model, in accordance with one or more embodiments;

[0017] FIG. **11** illustrates the diffusion processes used to train the diffusion model, in accordance with one or more embodiments;

[0018] FIG. **12** illustrates a schematic diagram of the summary page generation system in accordance with one or more embodiments;

[0019] FIG. **13** illustrates a flowchart of a series of acts in a method of generating a multimedia summary page using a document in accordance with one or more embodiments; and

[0020] FIG. **14** illustrates a block diagram of an exemplary computing device in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0021] One or more embodiments of the present disclosure include a summary page generation system that generates a multimedia summary page using a text document. One conventional approach involves manually creating a multimedia summary page, which includes summarizing text content of a document, positioning the text content, selecting text properties (e.g., a font style, a text size, and a text color), and selecting a background of the summary page. Other conventional approaches involve manipulating a summary page template. For example, the summary page template can include predefined text properties and/or predetermined backgrounds. Such conventional approaches are laborious and depend on the artistic talent of the user and/or the user's

experience using digital tools of a digital workplace. Additionally, such conventional approaches are limited to the template options available in a template library, which may be unfit or insufficient for a particular summary page. Further, such conventional approaches do not leverage the content of a document to automatically generate multimedia summary pages.

[0022] To address these and other deficiencies in conventional systems, the summary page generation system of the present disclosure automatically generates a diverse set of visually appealing and semantically relevant multimedia summary pages, with limited user input. The summary page generation system automatically extracts text content from a document. The text of the multimedia summary page is based on the extracted text from the document. The summary page generation system leverages generative AI to generate a semantically relevant image associated with the document. The image and text of the multimedia summary page are harmonized using one of two subsystems that are configured to arrange the text content of the summary page. A first subsystem iteratively harmonizes properties of the text content of the multimedia summary page on generated background imagery, whereas the second subsystem generates a harmonized multimedia summary page using a text-to-image generative model.

[0023] Providing a diverse set of multimedia summary pages reduces computing resources such as power, memory, and bandwidth spent editing, creating, or otherwise adapting summary pages to a specific style for a specific document. For example, computing resources are not wasted by a user's attempt to manifest the user's creative efforts. Instead, a user simply provides the summary page generation system with a document and receives one or more multimedia summary pages. Accordingly, computing resources such as power, memory, and bandwidth are preserved as the time associated with creating summary pages is reduced.

[0024] FIG. **1** illustrates a diagram of a process of generating summary pages based on documents, in accordance with one or more embodiments. In some embodiments, the summary page generation system **100** may be implemented as part of a natural language processing (NLP) suite of software. In some embodiments, a user may access the summary page generation system **100** via a client application executing on their computing device (e.g., a desktop, laptop, mobile device, etc.). In some embodiments, the client application (or "app") may be an application provided by the summary page generation system **100** (or a service provider corresponding to the NLP system or other entity). Additionally, or alternatively, the user may access the summary page generation system **100** via a browser-based application executing in a web browser installed on the user's computing device. Additionally, or alternatively, the summary page generation system **100** may be implemented entirely or in part on the user's computing device.

[0025] In some embodiments, the summary page generation system **100** is initiated to, for example, generate a summary page from a document. The summary page can be a multimedia synopsis of the document using text to summarize content of the document and generated background imagery to set the tone of the document and/or visually convey the synopsis of the document.

[0026] The summary page generation system **100** includes a Socratic model that leverages multimodal pretrained models. A pretrained model is a machine learning model that has been trained to perform one or more domain-neutral tasks using domain-neutral datasets. Domain-neutral datasets are widely available datasets, and domain-neutral tasks include domain-neutral text generation tasks, domain-neutral text summarization tasks, and the like. One or more pretrained models are configured to generate and/or extract text associated with the document **104**. One or more other pretrained models are configured to generate images associated with the extracted and/or generated text. As described herein, the one or more pretrained models can be fine-tuned.

[0027] As shown at numeral **1**, the input **102** is provided to the summary page generation system **100** to initiate the process of generating a summary page. The input **102** includes the document **104**, which may be a report, an assignment, a proposal, a thesis, or any other document including one or more pages of text (e.g., a text heavy document). A user provides the document **104** to be processed by the summary page generation system **100** by uploading the document **104** to the summary page generation system **100** directly or uploading the document **104** to a cloud-based storage location (or other Internet-accessible storage location) and providing a reference (e.g., a URL, URI, or other reference) to the document **104** to the summary page generation system **100**.

[0028] In some embodiments, the input **102** can include a user selection. Responsive to initiating the summary page generation system **100**, a user may be prompted with a type of summary page generation. For example, a first type of summary page generation is a balanced summary page generation that balances a text layout and generated background imagery. The first type of summary page generation is described herein as the manual harmonization subsystem. A second type of summary page generation is an image-focused summary page generation. The image-focused summary page generation generates background imagery around a text layout. The second type of summary page generation is described herein as the automatic harmonization subsystem.

[0029] In some embodiments, the user selection is optional because a default subsystem is executed (e.g., either the manual harmonization subsystem or the automatic harmonization subsystem described herein). For example, the default subsystem is triggered responsive to a user identity, based on matching the user identity to a stored user profile. User identities can include a username, a user number, an IP address, and the like. The stored user profile includes an indication of a subsystem (e.g., either the manual harmonization subsystem or the automatic harmonization subsystem) to be executed responsive to the corresponding user identity.

[0030] At numeral **2**, the document summarizer model **106** generates a text summary based on the text document. In some embodiments, to generate the text summary by calling one or more application programming interfaces (APIs). An API refers to an interface and/or communication protocol between a client and a server, such that if the client makes a request in a predefined format, the client should receive a response in a specific format or initiate a defined action (e.g., generate a test summary from document **104**). In other embodiments, the document summarizer model **106** is a language model configured to summarize text. For example, the document summarizer model **106** can summarize the contents of document **104** responsive to receiving a prompt

(e.g., a natural language instruction) instructing the document summarizer model **106** to extract and/or summarize the text in the document **104**.

[0031] A neural network may include a machine-learning model that can be tuned (e.g., trained) based on training input to approximate unknown functions. In particular, a neural network can include a model of interconnected digital neurons that communicate and learn to approximate complex functions and generate outputs based on a plurality of inputs provided to the model. For instance, the neural network includes one or more machine learning algorithms. In other words, a neural network is an algorithm that implements deep learning techniques, i.e., machine learning that utilizes a set of algorithms to attempt to model high-level abstractions in data. Additional details with respect to the use of neural networks within the summary page generation system **100** are discussed herein.

[0032] In some embodiments, the document summarizer model **106** truncates the text summary using a truncation operation and/or further summarizes the text summary. Iteratively summarizing the contents of the document **104** preserves the salient characteristics of the document **104** while trimming the content of the document **104**. In some embodiments, the document summarizer model **106** truncates the summarized text to a predetermined number of tokens, where the predetermined number of tokens is a user-configurable parameter.

[0033] Also at numeral **2**, the document summarizer model **106** generates a structured representation of the text summary. The structured representation is an ordered arrangement of text that is included in the document summary page **114** (e.g., summary page content). Structured representations are described in more detail herein.

[0034] In some embodiments, the document summarizer model **106** uses a structured representation prompt to generate the structured representation. For example, the document summarizer model **106** generates the structured representation prompt and subsequently uses the structured representation prompt to generate the structured representation. An example of the structured representation prompt is included in Table 1 below.

TABLE 1

Example Structured Representation Prompt
[[SUMMARIZED TEXT FROM DOCUMENT 104]] Generate a summary page for the input document on the above truncated extracted text in HTML5. The name of the author should be on a new line as mentioned in the input document, do not make up the author's name if not present.

[0035] As illustrated in Table 1 above, the prompt includes instructions to generate a structured representation of summary page content using the text summary. It should be appreciated that the document summarizer model **106** can use any prompting technique to generate the structured representation of the summary page content.

[0036] At numeral **3**, the prompt generator **108** generates the image generation prompt. In some embodiments, the prompt generator **108** generates a prompt to generate the image generation prompt and subsequently generates the image generation prompt. For example, the prompt generator **108** is a language model configured to generate prompts.

[0037] The prompt to generate the image generation prompt is based on the text summary and the structured representation of the text summary received from the document summarizer model **106**. The inclusion of both the text summary and structured representation as inputs to the prompt generator **108** improve the likelihood that the image generation prompt generated by the prompt generator **108** will be used by the diffusion system **112** to generate a relevant summary page (e.g., document summary page **114**). An image generation prompt based on only a text summary or the structed representation may result in the diffusion system **112** generating an irrelevant background image used as part of the document summary page **114**.

[0038] An example prompt used to generate the image generation prompt based on both the summarized text and the structured representation is illustrated in Table 2 below:

TABLE 2

Example Prompt used to generate an Image Generation Prompt
[[STRUCTURED REPRESENTATION IN HTML FORMAT]] [[SUMMARIZED TEXT FROM DOCUMENT 104]] Generate the most important theme for a background image using the above content.

[0039] It should be appreciated that Table 2 provides one example of a prompt used to generate an image generation prompt. Other prompt engineering techniques can be applied to generate the image generation prompt. For example, the prompt generator **108** can be instructed to generate the prompt used for the image generation prompt using chain of thought reasoning, which instructs the prompt generator **108** to explain the logic behind the generated image generation prompt.

[0040] Using the prompt, the prompt generator **108** generates an image generation prompt. Examples of the prompts generated by the prompt generator **108** are included in Table 3 below:

TABLE 3

Examples of Image Generation Prompts for a document about the global growth of Entity A, which is a company in the information and technology sector.
Example 1: A visualization of a global network with interconnected nodes representing a peer- to-peer system of Entity A. Example 2: A modern abstract image featuring neon colors and geometric shapes representing the nature of a digital word

[0041] As indicated in Table 3, the prompt generator **108** generated two diverse prompts given the same input document **104** and the same prompt illustrated using table 2. The diverse prompts encourage a diverse set of document summary pages **114**. In some embodiments, the prompt generator **108** is configured to generate an image generation prompt without first generating a prompt to generate the image generation prompt. That is, the prompt generator **108** generates a single prompt (e.g., the image generation prompt) using both the text summary and the structured representation.

[0042] While the document summarizer model **106** is shown as a separate model from the prompt generator **108**, in some embodiments, the operations of the summarizer

model **106** and the prompt generator **108** are performed using a single model. For example, a single language model can perform the operations of both the document summarizer model **106** and the prompt generator **108**.

[0043] At numeral **4**, the diffusion system **112** receives the structured representation (e.g., determined by the document summarizer model **106**) and the image generation prompt (e.g., determined by the prompt generator **108**). As described herein, the structured representation and the image generation prompt allow the diffusion system **112** to determine the visual and textual components of the document summary page **114**. In operation, the diffusion model **110** generates background imagery of the document summary page **114** using the image generation prompt, and a portion of the text summary is placed within the document summary page **114** based on the structured representation. The diffusion system **112** is described in more detail herein.

[0044] At numeral **5**, the diffusion system **112** outputs one or more multimedia summary documents (e.g., document summary page **114**). Examples of document summary pages **114** can include multimedia cover pages, infographics, flyers, posters, etc. In some embodiments, for a single input document **104**, a predetermined number of document summary pages **114** are generated. Accordingly, a user is presented with a set of diverse document summary pages **114**. The generated document summary page **114** can be the first page of document **104**, conveying multimedia information such as a summary of the document **104** content, the document **104** theme, the title, the subtitle, author details, and a semantically coherent visual background.

[0045] FIG. **2** illustrates two examples of structured representations of the summary page content generated by the document summarizer model, in accordance with one or more embodiments. For instance, in example **202**, the document summarizer model **106** (including a language model, for instance) created an HTML structured representation that does not include an author or any entity information. The document summarizer model **106** created the structured representation without this information because author related information and/or entity relation information (e.g., company information/organization information) was absent from the input document (such as document **104**). The document summarizer model **106** did not hallucinate and inject author information and/or entity information in the structured representation. Additionally, as shown in example **202**, the document summarizer model **106** summarized the input document **104** to create a concise title and subtitle. The title is tagged using heading tags to indicate, for instance, that the text associated with the heading tag should be rendered in a large font size. In contrast, the subtitle is tagged using paragraph tags to indicate, for instance, that the text associated with the paragraph tag should be rendered in a font size smaller than the font size associated with the heading tag. Accordingly, each tag of the structured document provides information related to the corresponding text style (e.g., a title or a subtitle, rendered using larger or smaller font respectively). Additionally, each tag of the structured document provides information related to the reading order. For example, the information associated with heading tags is prioritized on a page such that the information associated with the heading tags is read before the information associated with paragraph tags. In example **204**, the document summarizer model **106** generated the structured representation of the summary page content with author information and title information (as indicated using heading and paragraph tags respectively).

[0046] FIG. **3** illustrates a diagram of a process of the diffusion system, in accordance with one or more embodiments. As noted above, in some embodiments, the inputs to the summary page generation system **100** can include a user selection. For example, responsive to initiating the summary page generation system **100**, the user may be prompted with a type of summary page generation. Selecting the first type of summary page generation deploys the manual harmonization subsystem **314** which balances a text layout and generated background imagery to create the document summary page **114**. Selecting the second type of summary page generation deploys the automatic harmonization subsystem **316** which is an image-focused summary page generation that generates background imagery around a text layout to create the document summary page **114**.

[0047] The user selection of a particular type of summary page generation is passed to the routing manager **302**. If the user selects the first style of summary page generation, the routing manager **302** passes at least the structured representation and the image generation prompt to the manual harmonization subsystem **314**. If the user selects the second style of summary page generation, the routing manager **302** passes at least the structured representation and the image generation prompt to the automatic harmonization subsystem **316**.

[0048] The manual harmonization subsystem **314** iteratively determines an optimal layout of text objects included in the document summary page **114**. In operation, the manual harmonization subsystem **314** starts with a generated background and arranges text on the background. The generated background is determined using the image generation prompt and the diffusion model **110** model. In some embodiments, the diffusion model is any pretrained text-to-image generative model. The arrangement of text includes iteratively arranging the position of text, and selecting a font style of the text, a font color of the text, and a font size of the text. The generated summary page is rendered using a structured object representation of the optimal arrangement of text and the generated image background.

[0049] In contrast, the automatic harmonization subsystem **316** starts with an arrangement of text (e.g., a text position, a font style associated with the text, and a font size associated with the text) and generates a background for the arranged text using the diffusion model **110**. In some embodiments, the diffusion model **110** is a fine-tuned diffusion model that generates the document summary page **114** as a single object (e.g., an arrangement of text and a generated background). The fine-tuned diffusion model learns to automatically harmonize summary pages in the pixel space. Unlike the manual harmonization subsystem **314**, in which the document summary page **114** includes arranged text objects that can be represented using a structured object representation, the automatic harmonization subsystem **316** generates the document summary page **114** by setting a value of each pixel. In other embodiments, the automatic harmonization subsystem **316** uses a fine-tuned diffusion model to generate a background based on the arrangement of text, and subsequently superimposes the arrangement of text and the generated background to generate the document summary page **114**.

[0050] FIG. **4** illustrates examples of diverse multimedia summary documents determined using the summary page

generation system, in accordance with one or more embodiments. FIG. **4** illustrates two sets of summary pages, where each set of summary pages is associated with a single input. As shown, the first set of summary pages **402** is a different style from the second set of summary pages **404**. As described herein, the diffusion system **112** can include two types of summary page generation subsystems (e.g., the manual harmonization subsystem **314** or the automatic harmonization subsystem **316**). The manual harmonization subsystem **314** is the type of summary page generation that balances text layout and generated background imagery. As described herein, the manual harmonization subsystem **314** generates one or more summary pages **402** as arrangements of multiple text objects on a generated background. The manual harmonization subsystem **314** iteratively determines the optimal layout of the multiple text objects.

[0051] In contrast, the automatic harmonization subsystem **316** is the type of summary page generation that is image-focused. As described herein, the automatic harmonization subsystem **316** uses a diffusion model to generate one or more summary pages **404** as a single object. In other embodiments, the diffusion model generates a background associated with a control (e.g., a text canvas) and superimposes the control with the generated background. In operation, the automatic harmonization subsystem **316** generates background imagery around a text layout.

[0052] FIG. **5** illustrates a diagram of a process of the manual harmonization subsystem, in accordance with one or more embodiments. As described herein, a first type of summary page generation deploys the manual harmonization subsystem **314** which balances a text layout and generated background imagery to create the document summary page **114**. Using the manual harmonization subsystem **314**, the position of text elements identified from the structured representation is determined on a generated background image. Additionally, font properties of the text elements (including font style, font color, and font size) are determined for the positioned text elements. For ease of illustration, only the manual harmonization subsystem **314** is illustrated in the diffusion system **112**. However, in some embodiments, both the manual harmonization subsystem **314** and the automatic harmonization subsystem **316** are included in the diffusion system **112**.

[0053] The text-to-image generative model **502** receives the image generation prompt to determine a background image for the summary page. The text-to-image generative model **502** may be any pretrained generative AI module configured to generate an image using the image generation prompt. For example, the text-to-image generative model **502** can be a diffusion model. Example operations of diffusion models are described herein. The image generation prompt instructs the text-to-image generative model **502** to generate a background image that is invoked by document **104**.

[0054] The context manager **504** places text elements on the generated background image (e.g., determined by the text-to-image generative model **502**). In other words, the arrangement of the text elements is dependent on the generated background image. Each text element is extracted from the structured representation. For example, text elements include the text associated with tags of the structured representation. In operation, the context manager **504** performs constrained layout generation using the placement manager **508** and the font manager **516**.

[0055] The placement manager **508** is a model (e.g., a machine learning model) that performs constrained layout generation to place text elements on the generated background image. In operation, the placement manager **508** uses a genetic algorithm to iteratively optimize a layout by minimizing an energy function. A genetic algorithm is an algorithm based on competition to improve each generation of a solution (e.g., a generated layout) until an optimal solution is reached (or a number of training iterations of have been satisfied).

[0056] In some embodiments, the energy function used in the genetic algorithm is four dimensional, including a visual saliency dimension, an alignment dimension, an overlap dimension, and a reading order dimension. Each of the dimensions of the energy function represent a constraint associated with the text element placement. For example, one constraint associated with the constrained layout generation includes selecting text element locations that do not obscure salient regions of the generated background (e.g., a saliency constraint representing the visual saliency dimension). Another constraint associated with the constrained layout generation includes aligning text elements (e.g., an alignment constraint representing the alignment dimension). Yet another constraint associated with the constrained layout generation includes placing text elements in non-overlapping regions of the generated background image. In other words, the text elements are placed in open regions of the generated background image (e.g., an overlap constraint representing the overlap dimension). Another constraint associated with the constrained layout generation includes arranging text elements in a reading order such that higher priority text elements are read before lower priority text elements (e.g., a title is read before a subtitle).

[0057] The initialization manager **510** generates an initial set of candidate layouts. For each candidate layout K, the initialization manager **510** randomizes an initial coordinate position of each text element. The number of text elements placed by the initialization manager **510** is set to the number of text elements associated with tags in the structured representation. Each text element is associated with a bounding box with a center (x_c, y_c). In some embodiments, the top left corner of the bounding box is the origin (0,0).

[0058] The width and height of the bounding box depend on the text element. For example, the average number of words of the text element and the height of the text element depends on the HTML tag of the text element, obtained from the structured representation. Accordingly, a given text element has the property (t_i, l_i), where t_i is the text of the i^{th} text element and l_i is the corresponding HTML tag of the i^{th} text element. The initialization manager **510** determines the width of the text element and the height of the text element according to Equation (1) below:

$$h_{boundingbox} = D_h[l] \times \left[\frac{len(t)}{avg.\ \text{number of words per line for } l} \right] \qquad (1)$$

$$w_{boundingbox} = D_w[l]$$

[0059] In Equation (1) above, D_h is a dictionary that maps scalar values and D_w is a dictionary that maps scalar values. Accordingly, the size of the bounding box for a text element has a height $h_{boundingbox}$ and a width $w_{boundingbox}$.

[0060] In some embodiments, the randomized initial coordinate position of each text element is based on the dimension of the background image such that the generated layout can be determined for backgrounds of various sizes. For example, the initialization manager **510** randomizes the center of each bounding box (x_c, y_c) to be a coordinate within the range illustrated in Equation (2) below:

$$\frac{w_{background}}{2} \le x_c \le 1 - \frac{w_{background}}{2} \text{ and } \frac{h_{background}}{2} \le y_c \le 1 - \frac{h_{background}}{2} \quad (2)$$
$$w_{background}, h_{background}, x_c, y_c \in [0,1]$$

[0061] As shown in Equation (2) above, the center coordinate x_c of each bounding box associated with a text element is expressed as a fraction of the width of the generated background image, and the center coordinate y_c is expressed as a fraction of the height of the generated background image. In some embodiments, the initialization manager **510** further constraints the randomized initial coordinate position of each text element such that each element is not placed in a position exceeding the bounds of the generated background image. In other words, the initialization manager **510** checks that the width $w_{boundingbox}$ and height $h_{boundingbox}$ of a text element lay within the bounds of the generated image (e.g., the width and heigh of the generated images) based on the initialized center of the bounding box of each element (x_c, y_c).

[0062] In some embodiments, the initialization manager **510** checks that the width $w_{boundingbox}$ and height $h_{boundingbox}$ of the text element lays within the bounds of a predetermined margin. For example, using the bounds of the generated image (e.g., the width and heigh of the generated image), the initialization manager **510** accounts for a margin m_w from the width of the background image (e.g., the left and/or right sides of the background image) and/or margin m_h from the height of the background image (e.g. the top and/or bottom sides of the background image) to prevent a text element from being initialized at the edge of the background image. Accordingly, the center of each bounding box (x_c, y_c) is randomly selected within the range illustrated in Equation (3) below:

$$\frac{w_{background}}{2} + m_w \le x_c \le 1 - \frac{w_{background}}{2} - m_w \quad (3)$$
$$\frac{h_{background}}{2} + m_h \le y_c \le 1 - \frac{h_{background}}{2} - m_h$$

[0063] In operation, the initialization manager **510** initializes K candidate layouts, each including randomly positioned text elements determined from the structured representation. As described above, a structured representation can be expressed in terms of text elements according to $\ell = \{(t_1, l_1), (t_2, l_2), \ldots, (t_n, l_n)\}$, where t_i is the text of the i^{th} element and l_i is the corresponding HTML tag. The structured representation ℓ includes an implicit reading order. In other words, each of the HTML tags of the structured representation implicitly orders the text elements. For example, HTML tag l_1, which may be a heading HTML tag, indicates that the text element t_1 should be arranged in a higher priority reading order than the text element t_2 associated with HTML tag l_2, which may be a paragraph HTML tag.

[0064] Using the structured representation and the randomly initialized location of each text element, the initialization manager **510** generates K candidate layouts according to Equation (4) below:

$$\mathcal{L} = \{L_1, L_2, \ldots, L_K\} \quad (4)$$
$$L_i = \{(x_1^i, y_1^i, w(t_1, l_1), h(t_1, l_1)), (x_2^i, y_2^i, w(t_2, l_2), h(t_2, l_2)), \ldots, (x_n^i, y_n^i, w(t_n, l_n), h(t_n, l_n))\}$$

[0065] As described herein, the first two coordinates for each text element indicate a center coordinate of a bounding box (e.g., (x_c, y_c)), and the last two coordinates indicate the width and height of the bounding box that are dependent on the text element of the structured representation. The initialization manager **510** passes the set of candidate layouts $\mathcal{L}$ to the selection manager **512**.

[0066] The selection manager **512** selects one or more most viable layouts from the set of K candidate layouts determined by the initialization manager **510**. In operation, the selection manager **512** evaluates each candidate layout with respect to a set of constraints to determine the one or more most viable layouts from the set of K candidate layouts. The extent to which a candidate layout L_i satisfies a constraint is directly proportional to the negative energy function $\varepsilon(L_i)$. In other words, the energy function ε indicates how much the candidate layout L_i violates one or more constraints. The energy function is defined according to Equation (5) below:

$$\mathcal{E}(L_i) = w_{sal}\mathcal{L}_{sal} + w_{al}\mathcal{L}_{al} + w_{ov}\mathcal{L}_{ov} + w_{RO}\mathcal{L}_{RO} \quad (5)$$

[0067] In Equation (5) above, $\mathcal{L}_{sal}$ represents a saliency constraint and is a loss measure of how much the candidate layout L_i obscures the salient region of the background image. In operation, $\mathcal{L}_{sal}$ quantifies how much of the salient region of the background image is obscured by text elements given layout L_i. As shown in Equation (5) above, $\mathcal{L}_{sal}$ is weighted by w_{sal}. $\mathcal{L}_{al}$ is represents an alignment constraint and is a loss measure of alignment between text elements. In operation, $\mathcal{L}_{al}$ encourages alignment of a text element with respect to the left and right borders of the text element and/or with respect to the central axis of the text element. As shown in Equation (5) above, $\mathcal{L}_{al}$ is weighted by w_{al}. $\mathcal{L}_{ov}$ represents an overlap constraint and discourages text elements from overlapping. In operation, $\mathcal{L}_{ov}$ is a loss measure proportional to the sum of pairwise overlap between text elements. As shown in Equation (5) above, $\mathcal{L}_{ov}$ is weighted by w_{ov}. $\mathcal{L}_{RO}$ represents a reading order constraint and is a loss measuring the reading order of text elements in a candidate layout. As shown in Equation (5) above, $\mathcal{L}_{RO}$ is weighted by w_{RO}. In some embodiments, the weights w_{sal}, w_{al}, w_{ov}, and w_{RO} are predetermined. In some embodiments, the weights w_{sal}, w_{al}, w_{ov}, and w_{RO} are optional.

[0068] The selection manager **512** determines $\mathcal{L}_{sal}$ by determining a saliency map of the generated background image using any suitable technique, where the saliency map

indicates the degree of importance of each pixel of the generated background image with respect to an average user's visual perception of the generated background image. In some embodiments, the selection manager **512** determines the saliency map using images features extracted from the generated background image and a Gaussian pyramid. In some embodiments, the saliency map (H) represents an importance of each pixel of the generated background image using a value between 0 and 1, where values closer to 1 indicate more salient pixels. The inverse saliency map (H') represents pixels that are not salient, where H'=1−H. The selection manager **512** can also determine saliency maps and inverse saliency maps for candidate layouts such that H'_l represents the inverse saliency map associated with a candidate layout. For example, the selection manager **512** determines H'_l by masking the text elements of the candidate layout (e.g., represented by a pixel value of '1' for instance) to an array of pixels of the dimension of the generated background image. The array of pixels of the dimension of the generated background image (where the dimension of the background image includes a height and a width) can be initialized to a different pixel value (e.g., a blank array, where each pixel is set to the value '0'). Accordingly, the loss $\mathcal{L}_{sal}$ determined by the selection manager **512** using the inverse saliency maps is shown in Equation (6) below:

$$\mathcal{L}_{sal} = \frac{1}{N \times M} \sum_{i=1}^{N} \sum_{j=1}^{M} \max\left(H_{l(i,j)} - H'_{l(i,j)}, 0\right) \tag{6}$$

[0069] In Equation (6), N represents the height of the generated background image and M represents the width of the generated background image. As shown above, the selection manager **512** determines the mean of the element-wise difference between the inverse saliency map of the generated background image H' and the inverse saliency map of the candidate layout H'_l. The loss $\mathcal{L}_{sal}$ increases when pixels of text elements of the candidate layout overlap with salient regions of the background image. The loss $\mathcal{L}_{sal}$ ignores negative values of the element-wise difference because negative values indicate regions in the candidate layout without text elements that overlap with non-salient regions of the generated background image. Accordingly, candidate layouts that prioritize placement of text elements in non-salient regions of the background image are rewarded (or not penalized).

[0070] The selection manager **512** determines $\mathcal{L}_{al}$ using the bounding box of each text element in a candidate layout. For example, the selection manager **512** determines $\mathcal{L}_{al}$ according to Equation (7) below:

$$\mathcal{L}_{al} = \min(al_{center}, al_{left}, al_{right}) \tag{7}$$

where

$$al_{center} = \sum_{i=1}^{n} \Sigma_{j=i+1}^{n} \text{abs}(x_{ci} - x_{cj})$$

$$al_{left} = \sum_{i=1}^{n} \Sigma_{j=i+1}^{n} abs(x_{li} - x_{lj})$$

-continued

$$al_{right} = \sum_{i=1}^{n} \Sigma_{j=i+1}^{n} \text{abs}(x_{ri} - x_{rj})$$

[0071] In Equation (7) above, n represents the number of text elements in a candidate layout, x_l represents the left coordinate of the bounding box associated with a text element (e.g.,

$$x_c - \frac{\text{Width of the generated background image}}{2})$$

and x_r represents the right coordinate of the bounding box associated with the text element (e.g.,

$$x_c + \frac{\text{Width of the generated background image}}{2}).$$

Because a bounding box can align along any orientation (e.g., center, left, or right), the minimum alignment across the three orientations is used as the penalty term $\mathcal{L}_{al}$. It should be appreciated that alignment in the x-direction is considered, however, $\mathcal{L}_{al}$ can also be determined with respect to the y-direction.

[0072] The selection manager **512** determines $\mathcal{L}_{ov}$ by determining an amount of overlap between text elements. For example, $\mathcal{L}_{ov}$ can be determined using Equation (8) below:

$$x_{ov} = \sum_{i=1}^{n} \Sigma_{j=i+1}^{n} IoU(i, j) \tag{8}$$

[0073] In Equation (8) above, n represents the number of text elements in a candidate layout, and IoU(i,j) is the Intersection over Union used to determine whether text element i overlaps with text element j.

[0074] The selection manager **512** determines $\mathcal{L}_{RO}$ by comparing the position of each text element in a candidate layout to a position in the generated background image. For example, the position of each text element can be compared to the top-left corner of the generated background image. By comparing the position of each text element in a candidate layout to a position in the generated background image, the selection manager **512** creates a list $D_i = d_1^i, d_2^i, \ldots d_n^i$ where d_j^i is the distance of the jth element from the position in the generated background image (e.g., the top-left corner). As described herein, text elements are ordered by their corresponding HTML tags such that HTML tag l_i occurs before HTML tag l_k when i<k. Accordingly, text elements having a later position in the reading order will have a larger distance metric d. The selection manager **512** determines the reading order loss according to Equation (9) below:

$$\mathcal{L}_{RO} = \frac{\Sigma_{b \geq a} max(M[a,b]+\alpha, 0)}{|D_i| \times |D_i|} \tag{9}$$

-continued

where

$$\text{if } b \geq a \; M[a, b] = d_a^i - d_b^i$$

$$\text{else } M[a, b] = 0$$

[0075] In Equation (9) above, M represents an upper triangular matrix of text elements a and b, and α is a hyperparameter that encourages a margin between text elements. In some embodiments, α is predetermined. As shown, the loss $\mathcal{L}_{RO}$ compares a pair of text elements a and b. If a value of the matrix M is negative, then the distance of the later appearing text element (e.g., text element b) is greater than the former appearing text element (e.g., text element a). Accordingly, such a layout of text elements is consistent with the reading order and the two text elements are not penalized. In contrast, the loss $\mathcal{L}_{RO}$ penalizes pairs of text elements in which a value of the matrix M is positive.

[0076] The selection manager **512** creates two subsets of candidate layouts by comparing the value of the energy function of a candidate layout $\varepsilon(L_i)$ to a threshold. A first subset can include candidate layouts that satisfy the threshold. For example, the first subset can include candidate layouts associated with a low energy value (e.g., an energy value determined using the energy function that satisfies a threshold). In some embodiments, the candidate layouts of the first subset (e.g., candidate layouts associated with low energy values) includes a viable subset of candidate layouts. A second subset can include candidate layouts that do not satisfy the threshold. For example, the second subset can include candidate layouts associated with a high energy value (e.g., an energy value determined using the energy function that does not satisfy the threshold). In some embodiments, the candidate layouts of the second subset (e.g., candidate layouts associated with high energy values) includes a non-viable subset of candidate layouts. The subset of non-viable candidate layouts is passed to the reproduction manager **514**. The subset of viable candidate layouts is passed to the font manager **516**.

[0077] The reproduction manager **514** generates new candidate layouts using the subset of non-viable candidate layouts received from the selection manager **512**. For example, given two non-viable candidate layouts of the subset of non-viable candidate layouts (e.g., L_a and L_b), the reproduction manager **514** samples a new layout Lab from a distribution conditioned on non-viable candidates L_{aa} and L_b. Additionally or alternatively, the reproduction manager **514** can sample a new layout Lag from a distribution of a non-viable candidate layout L_a. The new candidate layouts are passed back to the selection manager **512** such that the selection manager can determine the energy value associated with the new candidate layouts. Using the above example, if the energy value associated with a new candidate layout satisfies the threshold (e.g., the energy value is low), the selection manager **512** adds the new candidate layout to the subset of viable candidate layouts. If the new candidate layout does not satisfy the threshold (e.g., the energy value is high), the selection manager **512** adds the new candidate layout to the subset of non-viable candidate layouts. The selection manager **512** recursively passes non-viable candidate layouts of the subset of non-viable candidate layouts to the reproduction manager **514**. The reproduction manager **514** recursively passes new candidate layouts, based on sampled non-viable candidate layouts, to the selection manager **512** to be evaluated (e.g., classified as a non-viable candidate layout and included in the subset of non-viable candidate layouts or classified as a viable candidate layout and included in the subset of viable candidate layouts) based on the energy value.

[0078] The font manager **516** determines font properties (e.g., such as font style, font color, and/or font size) of text elements of layouts in the subset of viable candidate layouts received from the selection manager **512**. The font style, font color, and/or font size are selected to contrast from the generated background image determined using text-to-image generative model **502**.

[0079] The font manager **516** selects a font style for a text element from a database of font styles using one or more classifiers. For example, a classifier generates a probability distribution over all font styles obtained from a database. The font manager **516** selects a font style for a text element based on a likelihood of the font style satisfying a threshold probability. In some embodiments, the font style with a likelihood satisfying a threshold probability indicates that the font style pairs well with the generated background image.

[0080] The font manager **516** selects a font size for a text element by determining the maximum font size possible, while ensuring that the text element fits within the bounding box. In some embodiments, the font manager **516** applies a binary search algorithm to iteratively increase or decrease the font size to be maximized with respect to the size of the bounding box, while still being constrained to the dimensions of the bounding box described herein (e.g., $h_{boundingbox}$ and $w_{boundingbox}$ described in Equation (1) above). In some embodiments, the binary search algorithm receives, as input, $h_{boundingbox}$ and $w_{boundingbox}$, the font style, and the text of the textual element.

[0081] The font manager **516** selects a font color for a text element. The selection of the font color is dependent on the region of the generated background image associated with the positioned text element. That is, the font manager **516** selects different font colors for different regions of the generated background image. Accordingly, the font manager **516** samples one or more colors of the generated background image at the location of the bounding box associated with the text element. Based on the sampled colors, the font manager **516** determines a dominant color of the generated background image at the location of the bounding box associated with the text element. For example, the font manager **516** applies the Modified Median Cut Quantization (MMCQ) algorithm to quantize pixels into a predetermined number of bins containing red, green blue (RGB) values. In operation, the font manager **516** quantizes pixels into d bins using a vector of length d by MMCQ $(X_{x1}^{x2}, _{y1}^{y2})$, where X is the generated background image, and (x_1, y_1) is a top coordinate of the bounding box (e.g., the top left coordinate of the bounding box including the text element) and (x_2, y_2) is a bottom coordinate of the bounding box (e.g., the bottom right coordinate of the bounding box including the text element). The dth bin with the largest number of pixels represents the dominant color (R_D, G_D, B_D).

[0082] Additionally, the font manager **516** extracts the color palette of the generated background image. For example, the font manager **516** samples random pixels to determine the color of the pixel in the generated background image. Additionally or alternatively, the font manager **516**

applies MMCQ to the generated background image to quantize pixels into p bins using a vector of length p.

[0083] Subsequently, the font manager **516** determines a color from the extracted color palette that has a greatest contrast ratio with the dominant color identified from the generated background image at the location of the bounding box. The color with the greatest contrast ratio is selected as the font color for the text element associated with the bounding box. In this manner, the selected font color is harmonious with the generated background image because it was selected from the color palette extracted from the generated background image. Further, the selected font color contrasts with the specific region of the generated background image with respect to the location of the text, identified via the bounding box.

[0084] In some embodiments, the font manager **516** selects one or more font properties in parallel. In other embodiments, the font manager **516** selects one or more font properties according to an order. For example, first the font manager **516** selects a font style, then the font manager **516** selects a font size, and lastly the font manager **516** selects a font color.

[0085] In some embodiments, the context manager **505** differentiates types of text elements, based on, for example, the text element tag identified in the structured representation (e.g., heading tags vs paragraph tags). For example, text elements associated with heading tags have similar properties (e.g., a first font size), and text elements associated with paragraph tags have similar properties (e.g., a second font size). In this manner, text elements associated with heading tags have different properties than text elements associated with paragraph tags.

[0086] The generated summary page, including a background generated using the text-to-image generative model **502** and text arranged, stylized, colored, and sized using the context manager **504**, is passed to the representation manager **518**. The representation manager **518** creates a structured object representation of the generated summary page, where the structured object representation includes attributes of the text (e.g., a font color, a font size, a font style, a text location). Such attributes are known because they were selected using the manual harmonization subsystem **314** (e.g., the text location determined by the selection manager **512**, the font style selected by the font manager **516**, the font size selected by the font manager **516**, the font color selected by the font manager **516**). The output of the diffusion system **112** executing the manual harmonization subsystem **314** is an intermediate document summary page that is a rendered version of the structured object representation. In some embodiments, the intermediate document summary page is the final document summary page (e.g., document summary page **114** described in FIG. **1**). In other embodiments, a user makes one or more edits to the intermediate document summary page using the edit manager **520**, as described below.

[0087] The edit manager **520** renders each object of the structured object representation (e.g., the intermediate document summary page) on an editable canvas. Accordingly, each object can be modified by a user. In operation, a user can edit the text (e.g., the content of the text and/or text properties), add new text, remove text, and the like, of the intermediate document summary page because the intermediate document summary page is rendered as a structured object representation. The edit manager **520** can receive user inputs indicating a modified font size, a modified font color, a modified font position, and/or a modified font style. Responsive to receiving such user inputs, the edit manager **520** updates the structured object representation, modifying displayed intermediate document summary page. After the user has finished editing the intermediate document summary page, the user obtains the document summary page **114** (e.g., a final version).

[0088] FIG. **6** illustrates a diagram of a process of the automatic harmonization subsystem, in accordance with one or more embodiments. As described herein, a second type of summary page generation deploys an automatic harmonization subsystem **316** which is an image-focused summary page generation to generate the document summary page **114**. The image-focused summary page generation generates background imagery around a text layout. Using the automatic harmonization subsystem **316**, a font style is selected and rendered onto an empty canvas. The canvas is then input to a fine-tuned diffusion model to generate a summary page. For ease of illustration, only the automatic harmonization subsystem **316** is illustrated in the diffusion system **112**. However, in some embodiments, both the manual harmonization subsystem **314** and the automatic harmonization subsystem **316** are included in the diffusion system **112**.

[0089] The text manager **602** renders text elements identified from the structured representation onto an empty canvas. The text manager **602** can perform operations similar to those of the placement manager **508** described with reference to FIG. **5**. For example, the text manager **602** can be a model (such as a machine learning model) that performs operation similar to those of the initialization manager **510**. As described above, each text element is associated with a bounding box, where the width and the height of the bounding box depend on the text element. For example, the average number of words of the text element and the height of the text element depends on the HTML tag of the text element, obtained from the structured representation. Accordingly, as discussed above, a given text element has the property (t_i, l_i), where t_i is the text of the i^{th} text element and l_i is the corresponding HTML tag of the i^{th} text element. The text manager **602** can determine the width of the text element and the height of the text element, according to Equation (1), reproduced below:

$$h_{boundingbox} = D_h[l] \times \left[\frac{len(t)}{avg.\ \text{number of words per line for } l} \right] \qquad (1)$$

$$w_{boundingbox} = D_w[l]$$

[0090] In Equation (1) above, D_h is a dictionary that maps scalar values and D_w is a dictionary that maps scalar values. Accordingly, the size of the bounding box for a text element has a height $h_{boundingbox}$ and a width $w_{boundingbox}$ Each bounding box has a center (x_c, y_c).

[0091] In some embodiments, the text manager **602** positions the center of each bounding box along a central axis of a blank canvas to create a text canvas, which is used as a control for the second text-to-image generative model **604** described herein. The central axis is the axis along the y-axis that bisects the blank canvas into two equal parts. In some embodiments, the text manager **602** randomly samples a coordinate position of a bounding box along the y-axis. In some embodiments, the sampled coordinate value is a value

within a range, where the range is predetermined based on a size of the summary page to be generated. For example, different summary page dimensions are associated with different predetermined ranges of coordinate values.

[0092] In some embodiments, the text manager **602** offsets bounding boxes associated with consecutive text elements. For example, the offset can be a spacing parameter that specifies the distance between consecutive text elements. In some embodiments, the offset is predetermined. In other embodiments, the offset is a user-configurable parameter (e.g., a received offset value as part of input **102**). In yet other embodiments, the offset is randomly sampled. Randomly sampling the offset value, the font style, and/or the position of the text elements on the y-axis increases the diversity of each generated summary page.

[0093] In some embodiments, the text manager **602** places the text element bounding boxes on an empty canvas similar to the operations performed by the placement manager **508** described in FIG. **5**. For example, the text manager **602** is a machine learning model that applies a genetic algorithm to iteratively optimize a layout by minimizing an energy function. For example, the energy function can be similar to the energy function described in Equation (6) above, with visual saliency modifications. An example energy function is shown below in Equation (10):

$$\mathcal{E}(L_i) = w_{al}\mathcal{L}_{al} + w_{ov}\mathcal{L}_{ov} + w_{RO}\mathcal{L}_{RO} \tag{10}$$

[0094] In Equation (10) above, the losses $\mathcal{L}_{al}$, $\mathcal{L}_{ov}$, and $\mathcal{L}_{RO}$ can be determined similarly to those described above. In other words, the text manager **602** can perform operations similar to those of the selection manager **512** described in FIG. **5**. The text arranged on the blank canvas becomes the controls passed to the second text-to-image generative model **604**.

[0095] The text manager **602** passes the control (e.g., the text canvas) to the second text-to-image generative model **604**. In some embodiments, the second text-to-image generative model **604** is a diffusion model such as a fine-tuned ControlNet model. ControlNet is a diffusion model configured to generate an image using the image generation prompt and a control. Example operations of diffusion models are described herein. Fine-tuning second text-to-image generative model **604** is described further herein.

[0096] In some embodiments, the output of the second text-to-image generative model **604** is the generated document summary page **114**. The document summary page **114** output by the second text-to-image generative model **604** respects the control (e.g., the location of the text identified using the text canvas) and includes a background image that is relevant given the image generation prompt. In other embodiments, the output of the second text-to-image generative model **604** is a generated background and the generated document summary page **114** is created by superimposing the control with the generated background. In these embodiments, a superposition manager **606** receives the generated background image and the control, superimposing the two objects to create the document summary page **114**. In some embodiments, the operations of the superposition manager **606** are performed by the second text-to-image generative model **604**.

[0097] FIG. **7** illustrates three examples of text canvases (or controls) determined using the text manager, in accordance with one or more embodiments. As described herein, a text canvas is used as a control for the second text-to-image generative model **604**. The second text-to-image generative model **604** generates a background that respects the control information (e.g., the position of text, the style of the font, the size of the font, etc.). In other words, the second text-to-image generative model **604** generates a background with respect to the control signal. Accordingly, the generated background appears visually appealing with respect to the text of the text canvas. Additionally, the generated background is relevant with respect to the document because of the received image generation prompt (e.g., based on the representation of the document and/or the structured representation of the document).

[0098] As described herein, the text manager **602** can randomly sample a font style for the text elements. As shown in examples **702**, the font style of each of the text elements within a text canvas is the same. For instance, the title font and the author font are the same font style. In other embodiments, the text manager **602** randomly samples a font style for each text element of the text canvas. For instance, the title font and the author font can be different font styles. Further, in the examples **702**, the text elements are positioned along the y-axis at a randomly sampled coordinate. For instance, the offset between the title and the author text is different in each of the three examples **702**. Additionally or alternatively, the location of the title text and the author text is randomly determined. As described herein, the text manager **602** can also iteratively place text in a constrained layout, based on text elements overlapping (e.g., $\mathcal{L}_{ov}$), text element alignment (e.g., $\mathcal{L}_{al}$) and text element reading order (e.g., $\mathcal{L}_{RO}$).

[0099] FIG. **8** illustrates an example process of generating training data to fine-tune a diffusion model, in accordance with one or more embodiments. Template **802** can be any content that includes a background and text. The template is used to fine-tune the diffusion model **110** of the automatic harmonization subsystem **316** (e.g., the second text-to-image generative model **604** such as ControlNet). As a result of the fine-tuning, the second text-to-image generative model **604** learns to generate an image while respecting the control. For example, the generated image background respects the arrangement of text conveys in the control. Accordingly, templates **802** should include features to be learned by diffusion model **110** (e.g., templates **802** should include text that is not overlapping, include text that is aligned, include text that does not obscure visually salient regions of the background, include text that contrasts with the background, etc.). In some embodiments, templates **802** are obtained from a template repository. In some embodiments, the structured object representation of each template **802** is known. Accordingly, the properties of the template **802** are known (e.g., the placement of text on the template **802**, the font size of text on the template **802**, the font style of text on the template **802**, the font color of text on the template **802**, and the like). In some embodiments, properties of the template **802** are obtained from one or more upstream processes.

[0100] The training data generator **804** decomposes the templates **802** into extracted text **806** (which is used as a control), extracted background **810**, and an image generation prompt associated with the template **802** (described herein

as a training image generation prompt). In embodiments where the structured object representation of the template is known, the training data generator **804** obtains extracted background **810** by setting all text properties to empty values. Accordingly, the training data generator **804** obtains an extracted background **810** without any rendered text.

[0101] As shown, the training data generator **804** includes an automatic captioning model. In some embodiments, the automatic captioning model is a bootstrapping language-image pre-training model (BLIP) **814**. The automatic captioning model (such as BLIP **814**) generates training image generation prompts **808**. While BLIP **814** is shown, any other descriptive model/visual understanding model can be deployed by the training data generator **804**. BLIP **814** is used to determine a prompt or a description of the images present in template **802**. In operation, the extracted background **810** is fed to BLIP **814** to determine the training image generation prompt **808**.

[0102] The training data generator **804** determines the control **806** associated with the template **802**. In some embodiments, the training data generator **804** uses any boundary detection algorithm to extract text **806** from the templates **802**. For example, the training data generator **804** may deploy any one or more optical character recognition algorithms to extract text (e.g., control **806**). In other embodiments, the training data generator **804** subtracts the extracted background **810** from the template **802** in pixel space to obtain the extracted text (or control **806**).

[0103] Accordingly, the training data generator **804** generates a triplet dataset used to fine-tune the second text-to-image generative model **604** t. The triplet dataset can include the extracted text (e.g., control **806**), the training image generation prompt **808** and either the extracted background **810** or the template **802**.

[0104] FIG. **9** illustrates an example process of training a text-to-image generative model, in accordance with one or more embodiments. Supervised learning is a method of training a machine learning model given input-output pairs. An input-output pair is an input with an associated known output (e.g., an expected output, a labeled output, a ground truth). The training manager **904** trains diffusion model **110** (or second text-to-image generative model **604** such as ControlNet) using known input-output pairs such that the diffusion model **110** learns how to predict known outputs given known inputs. It should be appreciated that while supervised learning is described, other training methods can be used by the training manager **904**. In the present disclosure, training diffusion model **110** includes fine-tuning the second text-to-image generative model **604** such as ControlNet.

[0105] As shown, the training manager **904** obtains the triplet dataset, including two training inputs **902** and a training output **908**. As described in FIG. **8**, the triplet dataset is determined by the training data generator **804**. The two training inputs **902** include a control **806** (e.g., extracted text) and an image generation prompt **808**. The training output **908** includes either the extracted background **810** or the template **802** associated with the training inputs **902**. For example, as described in FIG. **8**., the template **802** is used to obtain a corresponding control **806**, an image generation prompt **808**, and an extracted background **810**. In some embodiments, the triplet dataset includes the control **806** and the image generation prompt **808** (e.g., training inputs **902**) associated with the template **802** (e.g., the training output **908**). In other embodiments, the triplet dataset includes the control **806** and the image generation prompt **808** (e.g., training inputs **902**) and the associated extracted background **810** (e.g., the training output **908**).

[0106] During training, the diffusion model **110** learns to generate a layout based on the training output **908** used in the triplet dataset. For example, training diffusion model **110** using the extracted background **810** teaches the diffusion model **110** how to generate predicted output **916**. The learned layout mimics the layout of the extracted background **810**. For example, the predicted output **916** can include one or more images that highlight (or otherwise contrast with) the control **806** while still being a relevant generated image given the image generation prompt **808**. As described herein, the predicted output **916** is superimposed with the control **806** to generate the summary page.

[0107] Similarly, training the diffusion model **110** using the template **802** teaches the diffusion model **110** how to generate predicted output **926**. The learned layout of predicted output **926** mimics the layouts of the template **802**, such that the diffusion model **110** learns to generate a template that does not include overlapping text, aligns the text, the text contrasts the background, and the text does not obscure visually salient regions of the background image. Accordingly, the diffusion model **110** learns to generate a summary page by learning an arrangement of text elements over a generated background.

[0108] The diffusion model **110** uses the training inputs **902** to predict an output **906** by applying the current state of the diffusion model **110** to the training inputs **902**. The training manager **904** can compare the predicted outputs **906** to the known output (e.g., training output **908**) to determine an amount of error or differences. For example, the training manager **904** can determine one or more image features associated with the predicted output **906**, and one or more image features associated with the training output **908**. Subsequently, the training manager **904** can compare the similarity of the image features associated with the predicted output **906** and the image features associated with the training output **908**, using, for instance, cosine similarity or any other similarity metric. The dissimilarity between the image features (e.g., **1**-the similarity score determined using the similarity metric) can be expressed as the error.

[0109] The error (represented by error signal **910**) may be used to adjust the weights in the diffusion model **110** such that the diffusion model **110** changes (or learns) over time to generate a relatively accurate predicted output **906** using the triplet dataset (e.g., training inputs **902** and corresponding training output **908**). The diffusion model **110** may be trained using the backpropagation algorithm, for instance. The backpropagation algorithm operates by propagating the error signal **910**. The error signal **910** may be calculated each iteration (e.g., each pair of training inputs **902** and associated training output **908**), batch, and/or epoch and propagated through all of the algorithmic weights in the diffusion model **110** such that the algorithmic weights adapt based on the amount of error. The error is minimized using a loss function. Non-limiting examples of loss functions may include the square error function, the room mean square error function, and/or the cross-entropy error function.

[0110] The weighting coefficients of the diffusion model **110** may be tuned to reduce the amount of error thereby minimizing the differences between (or otherwise converging) the predicted output **906** and the training output **908**.

The diffusion model **110** may be trained until the error is within a certain threshold (or a threshold number of batches, epochs, or iterations have been reached).

[0111] FIG. **10** illustrates an example implementation of a diffusion model, in accordance with one or more embodiments. As described herein, any generative AI can be executed to generate an image related to visual text using the text-to-image generative model. In some embodiments, the text-to-image generative model is a generative AI model such as a diffusion model.

[0112] Generative AI involves predicting features for a given label. For example, given a label (or natural prompt description) "cat", the generative AI module determines the most likely features associated with a "cat." The features associated with a label are determined during training using a reverse diffusion process in which a noisy image is iteratively denoised to obtain an image. In operation, a function is determined that predicts the noise of latent space features associated with a label.

[0113] During training (e.g., using training manager **904** for instance), an image (e.g., an image of a cat) and a corresponding label (e.g., "cat") are used to teach the diffusion model features of a prompt (e.g., the label "cat"). As shown in FIG. **10**, an input image **1002** and a text input **1012** are transformed into latent space **1020** using an image encoder **1004** and a text encoder **1014** respectively. After the text encoder **1014** and image encoder **1004** have encoded text input **1012** and image input **1002** respectively, image features **1006** and text features **1008** are determined from the image input **1002** and text input **1012** accordingly. The latent space **1020** is a space in which unobserved features are determined such that relationships and other dependencies of such features can be learned. In some embodiments, the image encoder **1004** and/or text encoder **1014** are pretrained. In other embodiments, the image encoder **1004** and/or text encoder are trained jointly.

[0114] Once image features **1006** have been determined by the image encoder **1004**, a forward diffusion process **1016** is performed according to a fixed Markov chain to inject gaussian noise into the image features **1006**. The forward diffusion process **1016** is described in more detail herein. As a result of the forward diffusion process **1016**, a set of noisy image features **1010** are obtained.

[0115] The text features **1008** and noisy image features **1010** are algorithmically combined in one or more steps (e.g., iterations) of the reverse diffusion process **1026**. The reverse diffusion process **1026** is described in more detail herein. As a result of performing reverse diffusion, image features **1018** are determined, where such image features **1018** should be similar to image features **1006**. The image features **1018** are decoded using image decoder **1022** to predict image output **1024**. Similarity between image features **1006** and **1018** may be determined in any way. In some embodiments, instead of comparing similarity between image features, the similarity between images (e.g., image input **1002** and predicted image output **1024**) is determined in any way. The similarity between image features **1006** and **1018** and/or images **1002** and **1024** are used to adjust one or more parameters of the reverse diffusion process **1026**.

[0116] FIG. **11** illustrates the diffusion processes used to train the diffusion model, in accordance with one or more embodiments. The diffusion model may be implemented using any artificial intelligence/machine learning architecture in which the input dimensionality and the output dimensionality are the same. For example, the diffusion model may be implemented according to a u-net neural network architecture.

[0117] As described herein, a forward diffusion process adds noise over a series of steps (iterations t) according to a fixed Markov chain of diffusion. Subsequently, the reverse diffusion process removes noise to learn a reverse diffusion process to construct a desired image (based on the text input) from the noise. During deployment of the diffusion model, the reverse diffusion process is used in generative AI modules to generate images from input text. In some embodiments, an input image is not provided to the diffusion model.

[0118] The forward diffusion process **1016** starts at an input (e.g., feature X, indicated by **1102**). Each time step t (or iteration) up to a number of T iterations, noise is added to the feature X such that feature X_T indicated by **1110** is determined. As described herein, the features that are injected with noise are latent space features. If the noise injected at each step size is small, then the denoising performed during reverse diffusion process **1026** may be accurate. The noise added to the feature X can be described as a Markov chain where the distribution of noise injected at each time step depends on the previous time step. That is, the forward diffusion process **1016** can be represented mathematically $q(X_{1:T}|X_0)=\Pi_{t=1}^{T}q(X_t|X_{t-1})$.

[0119] The reverse diffusion process **1026** starts at a noisy input (e.g., noisy feature X_T indicated by **1110**). Each time step t, noise is removed from the features. The noise removed from the features can be described as a Markov chain where the noise removed at each time step is a product of noise removed between features at two iterations and a normal Gaussian noise distribution. That is, the reverse diffusion process **1126** can be represented mathematically as a joint probability of a sequence of samples in the Markov chain, where the marginal probability is multiplied by the product of conditional probabilities of the noise added at each iteration in the Markov chain. In other words, the reverse diffusion process **1026** is $p_{74}(X_{0:T})=p(X_t)\Pi_{t=1}^{T}p_\theta(X_{t-1}|X_t)$, where $p(X_t)=N(X_t; 0,1)$.

[0120] FIG. **12** illustrates a schematic diagram of a summary page generation system (e.g., "summary page generation system" described above) in accordance with one or more embodiments. As shown, the summary page generation system **1200** may include, but is not limited to, a user interface manager **1202**, a document summarizer **1204**, a prompt generator **1206**, a diffusion system **1208**, a neural network manager **1212**, a training manager **1218**, a training data generator **1226**, and a storage manager **1220**. The neural network manager **1212** includes a diffusion model **1210**.

[0121] As illustrated in FIG. **12**, the summary page generation system **1200** includes a user interface manager **1202**. For example, the user interface manager **1202** allows users to provide an input document to the summary page generation system **1200**. In some embodiments, the user interface manager **1202** provides a user interface through which the user can upload the input documents. For example, as discussed above, the text information is extracted and/or summarized using the document (e.g., author information, a title, a subtitle, a summary, etc.) to be used to generate the summary page. In some embodiments, the user interface enables the user to download the document from a local or remote storage location (e.g., by providing an address (e.g., a URL or other endpoint) associated with a document database or other document source).

[0122] Additionally, the user interface manager **1202** allows users to request the summary page generation system **1200** edit the generated summary page such as by changing the font style, font size, font color, and/or font position of text rendered on the generated summary page. In some embodiments, the user interface manager **1202** enables the user to view the resulting output generated summary page and/or request further edits to the generated summary page.

[0123] As illustrated in FIG. **12**, the summary page generation system **1200** includes a document summarizer **1204**. The document summarizer **1204** generates a text summary based on the text document. The document summarizer **1204** also generates a structured representation prompt and/or a structured representation using the text summary. In some embodiments, the document summarizer **1204** is a language model. In other embodiments, the document summarizer **1204** uses one or more API to perform the above-described operations.

[0124] In some embodiments, the document summarizer **1204** generates the structured representation based on a structured representation prompt that includes a text summary of the document. The structured representation prompt instructs the language model to generate the structured representation in an HTML format with tags associated with content of the document. The one or more tags associated with the content of the document can include a tag indicating a title of the document, a tag indicating the author of the document, a tag indicating a summary of the content of the document, and the like. Each tag of the structured document provides information related to the corresponding text style (e.g., a title or a subtitle, rendered using larger or smaller font respectively). For example, the structured representation can include paragraph tags and heading tags. Additionally, each tag of the structured document provides information related to the reading order. For example, the information associated with heading tags is prioritized on a page such that the information associated with the heading tags is read before the information associated with paragraph tags.

[0125] As illustrated in FIG. **12**, the summary page generation system **1200** includes a prompt generator **1206**. The prompt generator **1206** generates the image generation prompt and/or a prompt used to determine the image generation prompt. The image generation prompt is based on the structured representation of the document, and in some embodiments, the text summary of the document (e.g., from which the structured representation is based on).

[0126] As illustrated in FIG. **12**, the summary page generation system **1200** includes a diffusion system **1208**. The diffusion system **1208** can include one or two subsystems such as the manual harmonization subsystem and the automatic harmonization subsystem described herein. The manual harmonization subsystem and automatic harmonization subsystem each generate one or more document summary pages.

[0127] As described herein, the manual harmonization subsystem iteratively determines an optimal layout of text objects included in the generated document summary page. In operation, the manual harmonization subsystem starts with a generated background and arranges text on the background. The background is generated using a text-to-image generative model (such as diffusion model **1210**). The arrangement of text includes arranging the position of text, and selecting a font style of the text, a font color of the text, and a font size of the text.

[0128] The automatic harmonization subsystem starts with an arrangement of text (e.g., a text position, a font size, and a font style) and generates a background for the arranged text. In some embodiments, the automatic harmonization subsystem uses diffusion model **1210** to generate the document summary page as a single object (e.g., an arrangement of text and a generated background). In other embodiments, the automatic harmonization subsystem uses the diffusion model **1210** generate a background based on the arrangement of text, and subsequently superimposes the arrangement of text and the generated background to generate the document summary page.

[0129] As illustrated in FIG. **12**, the summary page genteration system **1200** also includes a neural network manager **1212**. Neural network manager **1212** may host a plurality of neural networks or other machine learning models, such as the diffusion model **1210**. The neural network manager **1212** may include an execution environment, libraries, and/or any other data needed to execute the machine learning models. In some embodiments, the neural network manager **1212** may be associated with dedicated software and/or hardware resources to execute the machine learning models. The structured representation and the image generation prompt received by diffusion model **1210** are used to create the multimedia document summary page. The multimedia document summary page includes a text component that uses a portion of text from the text summary and generated background imagery determined using diffusion model **1210**.

[0130] In some embodiments, the diffusion model **1210** can be any pretrained machine learning model configured to perform one or more natural language processing tasks. For example, when the diffusion model **1210** is deployed in the the manual harmonization subsystem, the diffusion model **1210** can be any text-to-image generative machine learning model. In these embodiments, the diffusion model **1210** receives an image generation prompt and generates background imagery. As described herein, the manual harmonization subsystem iteratively arranges text content on the generated image to generate the summary page.

[0131] In some embodiments, the diffusion model **1210** is a fine-tuned machine learning model such as ControlNet. For example, when the diffusion model **1210** is deployed responsive to the automatic harmonization subsystem, the diffusion model **1210** can be a fine-tuned text-to-image generative machine learning model such as ControlNet. In these embodiments, the diffusion model **1210** receives an image generation prompt and a control (e.g., a text canvas including text content to be included in the generated page summary) and generates a summary page. In some embodiments, a background is generated, and the background is superimposed with the text canvas to generate the summary page.

[0132] Although depicted in FIG. **12** as being hosted by a single neural network manager **1212**, in various embodiments the neural networks (e.g., the document summarizer **1204**, if implementing a language model, the prompt generator **1206**, if implementing a language model, and diffusion model **1210**) may be hosted in multiple neural network managers and/or as part of different components. For example, the neural networks can be hosted by their own

neural network manager, or other host environment, in which the respective neural networks execute, or the neural networks may be spread across multiple neural network managers depending on, e.g., the resource requirements of each neural network, etc.

[0133] As illustrated in FIG. **12**, the summary page generation system **1200** also includes a training data generator **1226**. The training data generator **1226** generates triplet training data including two training inputs and one training output. For example, a single instance of triplet training data includes a control and an image generation prompt (e.g., training inputs) associated with an extracted background (e.g., a training output). Another instance of triplet training data includes a control and an image generation prompt (e.g., training inputs) associated with template (e.g., a training output).

[0134] As illustrated in FIG. **12** the summary page generation system **1200** also includes training manager **1218**. The training manager **1218** can teach, guide, tune, and/or train one or more neural networks. In particular, the training manager **1218** can train a neural network based on a plurality of training data. For example, the diffusion model **1210** deployed by the automatic harmonization subsystem is fine-tuned to generate the summary page.

[0135] As illustrated in FIG. **12**, the summary page generation system **1200** also includes the storage manager **1220**. The storage manager **1220** maintains data for the summary page generation system **1200**. The storage manager **1220** can maintain data of any type, size, or kind as necessary to perform the functions of the summary page generation system **1200**. The storage manager **1220**, as shown in FIG. **12**, includes the training data **1222** and generated summary pages **1224**. In some embodiments, the training data **1222** includes the templates used to obtain the triplet training data, as described herein. In other embodiments, the training data **1222** includes the triplet training data (e.g., control data such as extracted text, image generation prompts, extracted backgrounds and/or templates). As described herein, the training data generator **1226** generates training data **1222** using, for instance, templates. For example, the training data generator **1226** decomposes the templates into extracted text (which is used as a control), an extracted background, and an image generation prompt associated with the template (described herein as a training image generation prompt) to generate triplet training data. The training data **1222** is utilized by the training manager **1218** to train one or more neural networks to generate a harmonized summary page. For example, as described herein, the training manager **1218** uses supervised learning to fine-tune a ControlNet diffusion model using the training data **1222**. The generated summary pages **1224** include one or more generated summary pages determined using the diffusion system **1208** (e.g., the manual harmonization subsystem and/or the automatic harmonization subsystem). For example, the storage manager **1220** stores such generated summary pages **1224** for subsequent processing and/or future use by the user.

[0136] Each of the components of the summary page generation system **1200** and their corresponding elements (as shown in FIG. **12**) may be in communication with one another using any suitable communication technologies. It will be recognized that although components and their corresponding elements are shown to be separate in FIG. **12**, any of and their corresponding elements may be combined into fewer components, such as into a single facility or module, divided into more components, or configured into different components as may serve a particular embodiment.

[0137] The components and their corresponding elements can comprise software, hardware, or both. For example, the components and their corresponding elements can comprise one or more instructions stored on a computer-readable storage medium and executable by processors of one or more computing devices. When executed by the one or more processors, the computer-executable instructions of the summary page generation system **1200** can cause a client device and/or a server device to perform the methods described herein. Alternatively, the components and their corresponding elements can comprise hardware, such as a special purpose processing device to perform a certain function or group of functions. Additionally, the components and their corresponding elements can comprise a combination of computer-executable instructions and hardware.

[0138] Furthermore, the components of the summary page generation system **1200** may, for example, be implemented as one or more stand-alone applications, as one or more modules of an application, as one or more plug-ins, as one or more library functions or functions that may be called by other applications, and/or as a cloud-computing model. Thus, the components of the summary page generation system **1200** may be implemented as a stand-alone application, such as a desktop or mobile application. Furthermore, the components of the summary page generation system **1200** may be implemented as one or more web-based applications hosted on a remote server. Alternatively, or additionally, the components of the summary page generation system **1200** may be implemented in a suite of mobile device applications or "apps."

[0139] As shown, the summary page generation system **1200** can be implemented as a single system. In other embodiments, the summary page generation system **1200** can be implemented in whole, or in part, across multiple systems. For example, one or more functions of the summary page generation system **1200** can be performed by one or more servers, and one or more functions of the summary page generation system **1200** can be performed by one or more client devices. The one or more servers and/or one or more client devices may generate, store, receive, and transmit any type of data used by the summary page generation system **1200**, as described herein.

[0140] In one implementation, the one or more client devices can include or implement at least a portion of the summary page generation system **1200**. In other implementations, the one or more servers can include or implement at least a portion of the summary page generation system **1200**. For instance, the summary page generation system **1200** can include an application running on the one or more servers or a portion of the summary page generation system **1200** can be downloaded from the one or more servers. Additionally or alternatively, the summary page generation system **1200** can include a web hosting application that allows the client device(s) to interact with content hosted at the one or more server(s).

[0141] For example, upon a client device accessing a webpage or other web application hosted at the one or more servers, in one or more embodiments, the one or more servers can provide access to the user interface manager **1202** stored at the one or more servers. Moreover, the client device can receive a request (i.e., via user input) to generate a summary page and provide the request to the one or more

servers. Upon receiving the request, the one or more servers can automatically perform the methods and processes described above to generate one or more diverse summary pages. In some embodiments, the request to the one or more servers includes a user input identifying a summary page generation method to be implemented (e.g., the manual harmonization subsystem or the automatic harmonization subsystem of the diffusion system **1208**). In other embodiments, the one or more servers performs a default summary page generation method (e.g., the manual harmonization subsystem or the automatic harmonization subsystem). In some embodiments, the one or more servers selects the default page generation method based on a user identity (e.g., a username, a user account, an IP address, etc.) associated with the client device and/or the user. The one or more servers can provide the one or more generated summary pages, to the client device for display to the user.

[0142] The server(s) and/or client device(s) may communicate using any communication platforms and technologies suitable for transporting data and/or communication signals, including any known communication technologies, devices, media, and protocols supportive of remote data communications, examples of which will be described in more detail below with respect to FIG. **14**. In some embodiments, the server(s) and/or client device(s) communicate via one or more networks. A network may include a single network or a collection of networks (such as the Internet, a corporate intranet, a virtual private network (VPN), a local area network (LAN), a wireless local network (WLAN), a cellular network, a wide area network (WAN), a metropolitan area network (MAN), or a combination of two or more such networks. The one or more networks will be discussed in more detail below with regard to FIG. **14**.

[0143] The server(s) may include one or more hardware servers (e.g., hosts), each with its own computing resources (e.g., processors, memory, disk space, networking bandwidth, etc.) which may be securely divided between multiple customers (e.g. client devices), each of which may host their own applications on the server(s). The client device(s) may include one or more personal computers, laptop computers, mobile devices, mobile phones, tablets, special purpose computers, TVs, or other computing devices, including computing devices described below with regard to FIG. **14**.

[0144] FIGS. **1-12**, the corresponding text, and the examples, provide a number of different systems and devices that allow a user to generate one or more summary pages using an input document. In addition to the foregoing, embodiments can also be described in terms of flowcharts comprising acts and steps in a method for accomplishing a particular result. For example, FIG. **13** illustrates a flowchart of an exemplary method in accordance with one or more embodiments. The method described in relation to FIG. **13** may be performed with fewer or more steps/acts or the steps/acts may be performed in differing orders. Additionally, the steps/acts described herein may be repeated or performed in parallel with one another or in parallel with different instances of the same or similar steps/acts.

[0145] FIG. **13** illustrates a flowchart **1300** of a series of acts in a method of generating a multimedia summary page using a document in accordance with one or more embodiments. In one or more embodiments, the method **1300** is performed in a digital medium environment that includes the summary page generation system **1200**. The method **1300** is intended to be illustrative of one or more methods in accordance with the present disclosure and is not intended to limit potential embodiments. Alternative embodiments can include additional, fewer, or different steps than those articulated in FIG. **13**.

[0146] As illustrated in FIG. **13**, the method **1300** includes an act **1302** of receiving a text document. The text document may be a report, an assignment, a proposal, a thesis, or any other document including one or more pages of text (e.g., a text heavy document).

[0147] As illustrated in FIG. **13**, the method **1300** includes an act **1304** of generating, by a document summarizer model, a text summary based on the text document and a structured representation of the text summary. The document summarizer model generates a text summary based on the text document. For example, the document summarizer model can summarize the contents of the text document responsive to receiving a prompt (e.g., a natural language instruction) instructing the document summarizer model to extract and/or summarize the text in the text document. The document summarizer model **106** also generates a structured representation of the text summary. The structured representation is an ordered arrangement of text that is included in the multimedia summary document. In some embodiments, the document summarizer model uses a structured representation prompt to generate the structured representation. For example, the document summarizer model generates the structured representation prompt and subsequently uses the structured representation prompt to generate the structured representation.

[0148] As illustrated in FIG. **13**, the method **1300** includes an act **1304** of generating, by a prompt generator, an image generation prompt based on the text summary and the structured representation of the text summary. The prompt generator generates the image generation prompt and/or generates a prompt to generate the image generation prompt. The prompt to generate the image generation prompt is based on the text summary and the structured representation of the text summary received from the document summarizer model. Accordingly, the image generation prompt is based on the text summary and the structured representation of the text summary received from the document summarizer model.

[0149] As illustrated in FIG. **13**, the method **1300** includes an act **1306** of generating, using a diffusion model and the image generation prompt, a multimedia summary document corresponding to the text document. The multimedia summary document includes a generated background imagery based on the text summary. The multimedia summary document also includes at least a portion of the text summary which is placed within the multimedia summary document based on the structured representation of the text summary. The structured representation and the image generation prompt allow the diffusion model to determine the visual and textual components of the multimedia summary document corresponding to the text document. In operation, the diffusion model generates background imagery of the multimedia summary document using the image generation prompt, and a portion of the text summary is placed within the multimedia summary document based on the structured representation.

[0150] Embodiments of the present disclosure may comprise or utilize a special purpose or general-purpose computer including computer hardware, such as, for example, one or more processors and system memory, as discussed in

greater detail below. Embodiments within the scope of the present disclosure also include physical and other computer-readable media for carrying or storing computer-executable instructions and/or data structures. In particular, one or more of the processes described herein may be implemented at least in part as instructions embodied in a non-transitory computer-readable medium and executable by one or more computing devices (e.g., any of the media content access devices described herein). In general, a processor (e.g., a microprocessor) receives instructions, from a non-transitory computer-readable medium, (e.g., a memory, etc.), and executes those instructions, thereby performing one or more processes, including one or more of the processes described herein.

[0151] Computer-readable media can be any available media that can be accessed by a general purpose or special purpose computer system. Computer-readable media that store computer-executable instructions are non-transitory computer-readable storage media (devices). Computer-readable media that carry computer-executable instructions are transmission media. Thus, by way of example, and not limitation, embodiments of the disclosure can comprise at least two distinctly different kinds of computer-readable media: non-transitory computer-readable storage media (devices) and transmission media.

[0152] Non-transitory computer-readable storage media (devices) includes RAM, ROM, EEPROM, CD-ROM, solid state drives ("SSDs") (e.g., based on RAM), Flash memory, phase-change memory ("PCM"), other types of memory, other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory storage medium which can be used to store desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer.

[0153] A "network" is defined as one or more data links that enable the transport of electronic data between computer systems and/or modules and/or other electronic devices. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer, the computer properly views the connection as a transmission medium. Transmissions media can include a network and/or data links which can be used to carry desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer. Combinations of the above should also be included within the scope of computer-readable media.

[0154] Further, upon reaching various computer system components, program code means in the form of computer-executable instructions or data structures can be transferred automatically from transmission media to non-transitory computer-readable storage media (devices) (or vice versa). For example, computer-executable instructions or data structures received over a network or data link can be buffered in RAM within a network interface module (e.g., a "NIC"), and then eventually transferred to computer system RAM and/or to less volatile computer storage media (devices) at a computer system. Thus, it should be understood that non-transitory computer-readable storage media (devices) can be included in computer system components that also (or even primarily) utilize transmission media.

[0155] Computer-executable instructions comprise, for example, instructions and data which, when executed at a processor, cause a general-purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. In some embodiments, computer-executable instructions are executed on a general-purpose computer to turn the general-purpose computer into a special purpose computer implementing elements of the disclosure. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, or even source code. Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the described features or acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

[0156] Those skilled in the art will appreciate that the disclosure may be practiced in network computing environments with many types of computer system configurations, including, personal computers, desktop computers, laptop computers, message processors, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, mobile telephones, PDAs, tablets, pagers, routers, switches, and the like. The disclosure may also be practiced in distributed system environments where local and remote computer systems, which are linked (either by hardwired data links, wireless data links, or by a combination of hardwired and wireless data links) through a network, both perform tasks. In a distributed system environment, program modules may be located in both local and remote memory storage devices.

[0157] Embodiments of the present disclosure can also be implemented in cloud computing environments. In this description, "cloud computing" is defined as a model for enabling on-demand network access to a shared pool of configurable computing resources. For example, cloud computing can be employed in the marketplace to offer ubiquitous and convenient on-demand access to the shared pool of configurable computing resources. The shared pool of configurable computing resources can be rapidly provisioned via virtualization and released with low management effort or service provider interaction, and then scaled accordingly.

[0158] A cloud-computing model can be composed of various characteristics such as, for example, on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service, and so forth. A cloud-computing model can also expose various service models, such as, for example, Software as a Service ("SaaS"), Platform as a Service ("PaaS"), and Infrastructure as a Service ("IaaS"). A cloud-computing model can also be deployed using different deployment models such as private cloud, community cloud, public cloud, hybrid cloud, and so forth. In this description and in the claims, a "cloud-computing environment" is an environment in which cloud computing is employed.

[0159] FIG. **14** illustrates, in block diagram form, an exemplary computing device **1400** that may be configured to perform one or more of the processes described above. One will appreciate that one or more computing devices such as the computing device **1400** may implement the summary page generation system. As shown by FIG. **14**, the computing device can comprise a processor **1402**, memory **1404**,

one or more communication interfaces **1406**, a storage device **1408**, and one or more I/O devices/interfaces **1410**. In certain embodiments, the computing device **1400** can include fewer or more components than those shown in FIG. **14**. Components of computing device **1400** shown in FIG. **14** will now be described in additional detail.

[0160] In particular embodiments, processor(s) **1402** includes hardware for executing instructions, such as those making up a computer program. As an example, and not by way of limitation, to execute instructions, processor(s) **1402** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **1404**, or a storage device **1408** and decode and execute them. In various embodiments, the processor(s) **1402** may include one or more central processing units (CPUs), graphics processing units (GPUs), field programmable gate arrays (FPGAs), systems on chip (SoC), or other processor(s) or combinations of processors.

[0161] The computing device **1400** includes memory **1404**, which is coupled to the processor(s) **1402**. The memory **1404** may be used for storing data, metadata, and programs for execution by the processor(s). The memory **1404** may include one or more of volatile and non-volatile memories, such as Random Access Memory ("RAM"), Read Only Memory ("ROM"), a solid state disk ("SSD"), Flash, Phase Change Memory ("PCM"), or other types of data storage. The memory **1404** may be internal or distributed memory.

[0162] The computing device **1400** can further include one or more communication interfaces **1406**. A communication interface **1406** can include hardware, software, or both. The communication interface **1406** can provide one or more interfaces for communication (such as, for example, packet-based communication) between the computing device and one or more other computing devices **1400** or one or more networks. As an example and not by way of limitation, communication interface **1406** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI. The computing device **1400** can further include a bus **1412**. The bus **1412** can comprise hardware, software, or both that couples components of computing device **1400** to each other.

[0163] The computing device **1400** includes a storage device **1408** includes storage for storing data or instructions. As an example, and not by way of limitation, storage device **1408** can comprise a non-transitory storage medium described above. The storage device **1408** may include a hard disk drive (HDD), flash memory, a Universal Serial Bus (USB) drive or a combination these or other storage devices. The computing device **1400** also includes one or more input or output ("I/O") devices/interfaces **1410**, which are provided to allow a user to provide input to (such as user strokes), receive output from, and otherwise transfer data to and from the computing device **1400**. These I/O devices/interfaces **1410** may include a mouse, keypad or a keyboard, a touch screen, camera, optical scanner, network interface, modem, other known I/O devices or a combination of such I/O devices/interfaces **1410**. The touch screen may be activated with a stylus or a finger.

[0164] The I/O devices/interfaces **1410** may include one or more devices for presenting output to a user, including, but not limited to, a graphics engine, a display (e.g., a display screen), one or more output drivers (e.g., display drivers), one or more audio speakers, and one or more audio drivers. In certain embodiments, I/O devices/interfaces **1410** is configured to provide graphical data to a display for presentation to a user. The graphical data may be representative of one or more graphical user interfaces and/or any other graphical content as may serve a particular implementation.

[0165] In the foregoing specification, embodiments have been described with reference to specific exemplary embodiments thereof. Various embodiments are described with reference to details discussed herein, and the accompanying drawings illustrate the various embodiments. The description above and drawings are illustrative of one or more embodiments and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of various embodiments.

[0166] Embodiments may include other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. For example, the methods described herein may be performed with less or more steps/acts or the steps/acts may be performed in differing orders. Additionally, the steps/acts described herein may be repeated or performed in parallel with one another or in parallel with different instances of the same or similar steps/acts. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes that come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0167] In the various embodiments described above, unless specifically noted otherwise, disjunctive language such as the phrase "at least one of A, B, or C," is intended to be understood to mean either A, B, or C, or any combination thereof (e.g., A, B, and/or C). As such, disjunctive language is not intended to, nor should it be understood to, imply that a given embodiment requires at least one of A, at least one of B, or at least one of C to each be present.

We claim:

1. A method comprising:

receiving a text document;

generating, by a document summarizer model, a text summary based on the text document and a structured representation of the text summary;

generating, by a prompt generator, an image generation prompt based on the text summary and the structured representation of the text summary; and

generating, using a diffusion model and the image generation prompt, a multimedia summary document corresponding to the text document, wherein the multimedia summary document includes a generated background imagery based on the text summary, and wherein the multimedia summary document includes at least a portion of the text summary which is placed within the multimedia summary document based on the structured representation of the text summary.

2. The method of claim **1**, wherein generating, using the diffusion model and the image generation prompt, the multimedia summary document corresponding to the text document, further comprises:

generating, by the diffusion model, the generated background imagery using the image generation prompt;

determining, using a genetic algorithm, a position of the portion of the text summary;

determining a font color of the portion of the text summary;

determining a font style of the portion of the text summary; and

determining a font size of the portion of the text summary.

3. The method of claim **2**, wherein the genetic algorithm minimizes an energy function including a visual saliency loss, an alignment loss, an overlap loss, and a reading order loss.

4. The method of claim **3**, wherein the reading order loss is based on an order of text elements included in the structured representation.

5. The method of claim **1**, wherein generating, using the diffusion model and the image generation prompt, the multimedia summary document corresponding to the text document, further comprises:

generating a canvas using the structured representation; and

generating, by the diffusion model, the multimedia summary document using the canvas and the image generation prompt.

6. The method of claim **5**, wherein the diffusion model is trained using a triplet dataset, wherein the triplet dataset includes a training canvas, a training prompt, and a training summary page.

7. The method of claim **5**, wherein the diffusion model is a ControlNet diffusion model.

8. The method of claim **1**, wherein the structured representation is an HTML format including tags corresponding to text content of the text document.

9. A non-transitory computer-readable medium storing executable instructions, which when executed by a processing device, cause the processing device to perform operations comprising:

receiving a text document;

generating, by a document summarizer model, a text summary based on the text document and a structured representation of the text summary;

generating, by a prompt generator, an image generation prompt based on the text summary and the structured representation of the text summary; and

generating, using a diffusion model and the image generation prompt, a multimedia summary document corresponding to the text document, wherein the multimedia summary document includes a generated background imagery based on the text summary, and wherein the multimedia summary document includes at least a portion of the text summary which is placed within the multimedia summary document based on the structured representation of the text summary.

10. The non-transitory computer-readable medium of claim **9**, wherein generating, using the diffusion model and the image generation prompt, the multimedia summary document corresponding to the text document, further comprises:

generating, by the diffusion model, the generated background imagery using the image generation prompt;

determining, using a genetic algorithm, a position of the portion of the text summary;

determining a font color of the portion of the text summary;

determining a font style of the portion of the text summary; and

determining a font size of the portion of the text summary.

11. The non-transitory computer-readable medium of claim **10**, wherein the genetic algorithm minimizes an energy function including a visual saliency loss, an alignment loss, an overlap loss, and a reading order loss.

12. The non-transitory computer-readable medium of claim **11**, wherein the reading order loss is based on an order of text elements included in the structured representation.

13. The non-transitory computer-readable medium of claim **9**, wherein generating, using the diffusion model and the image generation prompt, the multimedia summary document corresponding to the text document, further comprises:

generating a canvas using the structured representation; and

generating, by the diffusion model, the multimedia summary document using the canvas and the image generation prompt.

14. The non-transitory computer-readable medium of claim **13**, wherein the diffusion model is trained using a triplet dataset, wherein the triplet dataset includes a training canvas, a training prompt, and a training summary page.

15. A system comprising:

a memory component; and

a processing device coupled to the memory component, the processing device to perform operations comprising:

receiving a text document and a user selection of a summary page generation type;

generating, by a first machine learning model, a text summary based on the text document;

generating, by the first machine learning model, a structured representation based on the text summary;

generating, by the first machine learning model, an image generation prompt based on the text summary and the structured representation; and

generating, by a second machine learning model, a multimedia summary document corresponding to the text document, wherein the multimedia summary document is generated based on the summary page generation type, the image generation prompt, and the structured representation.

16. The system of claim **15**, further comprising:

determining, using a genetic algorithm, a position of a summary page text included in the multimedia summary document;

determining a font color of the summary page text;

determining a font style of the summary page text; and

determining a font size of the summary page text.

17. The system of claim **16**, wherein the genetic algorithm minimizes an energy function including a visual saliency loss, an alignment loss, an overlap loss, and a reading order loss.

18. The system of claim **17**, wherein the reading order loss is based on an order of text elements included in the structured representation.

19. The system of claim **15**, further comprising:

generating a canvas using a summary page text included in the multimedia summary document; and

generating, by the second machine learning model, the multimedia summary document using the canvas and the image generation prompt.

20. The system of claim **15**, wherein the second machine learning model is trained using a triplet dataset, wherein the triplet dataset includes a training canvas, a training prompt, and a training summary page.

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